Multi-Component Style Vectors with Norm-Preserving Steering for Personalized Text Generation

Consolidated Experimental Report: Ablations from Runs 1–5

Slok Tulsyan
Student ID: 12342080
slok@iitbhilai.ac.in

October 29, 2026

Abstract

This project reimplements StyleVector (Zhang et al., ACL 2025), which represents a user's writing style as a single mean-difference direction in an LLM's activation space, and evaluates two extensions: *multi-component* style vectors and *norm-preserving* (NP) steering. This report consolidates **five experimental runs** on three LaMP tasks (news headline, scholarly title and tweet paraphrasing generation) with LLaMA-2-7B-chat. Runs 1–4 are successive iterations that differ in sample size, hyper-parameter selection, output post-processing and controls. Run 5 is the final run with a refined pipeline. Together they cover the planned 2×2 ablation (single vs. multi-component $\times$ additive vs. NP), sweeps over k, layer and strength, and global-vector, shrinkage and random-direction controls. Scores are compared with paired bootstrap confidence intervals.

Runs 1–4 show that evaluation choices matter as much as the method: output cleaning alone raised baseline ROUGE-L by 14–51%, hyper-parameters selected on validation users favoured very small steering strengths, and in Run 4 the large LaMP-7 effect was partly a formatting effect. Of 18 comparisons of the paper's method against no steering, 3 were significantly positive, 1 significantly negative and 14 inconclusive; for the extensions, 1 of 18 multi-component comparisons and 2 of 42 norm-preservation comparisons were significantly better than their counterparts.

Run 5 (refined pipeline: cleaned outputs, repaired history construction, per-condition tuning and a shared-basis multi-component vector) reproduces the paper's gains on LaMP-5 (+21.8% ROUGE-L, +10.9% METEOR against the paper's +25.8% and +10.2%) and improves ROUGE-L on LaMP-4 and LaMP-7 (+7.0% and +17.5%). Against this reproduced baseline, the shared-basis vector is statistically indistinguishable from Mean Difference in all 6 paired comparisons, and the norm-preserving variants score lower than Mean Difference in 3 of 6 comparisons each. Per-user PCA (k=3) matches Mean Difference and is better on LaMP-4 METEOR. The full ablation therefore identifies Mean Difference steering with tuned layer and strength as the strongest configuration tested.

Contents

1 Introduction and scope **3**

2 Method recap and evaluation protocol **3**
2.1 StyleVector baseline . . . 3
2.2 Proposed extension . . . 3
2.3 The 2×2 ablation and controls . . . 4

3 Overview of the five runs **4**

4 Run 1: first end-term attempt **5**

5 Run 2: validation-tuned, cleaned, with controls **6**

6 Run 3: larger validation and test sets, random control **9**

7 Run 4: k, layer and strength sweeps **10**
7.1 Diagnosing the LaMP-7 gain . . . 12
7.2 Layer and strength sweeps . . . 13

8 Run 5: corrected pipeline (final run) **15**
8.1 What changed and what was available . . . 15
8.2 Selected settings . . . 16
8.3 Results . . . 17
8.4 Paired comparisons against Mean Difference . . . 18
8.5 Findings . . . 19

9 Cross-run synthesis **19**
9.1 The 2×2 ablation across runs . . . 19
9.2 Effect of the paper's method and of the proposed extensions . . . 21
9.3 Comparison with the paper . . . 23
9.4 The output-format confound . . . 24
9.5 Is the gain personal? . . . 25
9.6 Strength selection and the role of NP . . . 25

10 Discussion **25**

11 Conclusion **26**

A Paired comparisons for every run **26**

B Additional original figures **28**

C Use of AI assistance **29**

1 Introduction and scope

This report documents every experimental attempt made so far for the final project, in the order they were run, together with an integrated analysis. It is intended to serve both as the ablation record for the final presentation and as the basis of the final report. Its scope is the four checkpoint archives plus the summary output of the final notebook run:

- **Run 1** (`checkpoints_endterm`): first end-term attempt, with hyper-parameters reused from the midterm and no output post-processing.
- **Run 2** (`checkpoints_v2`): adds validation-based tuning, output cleaning, a global-vector control, a shrinkage variant and paired statistics.
- **Run 3** (`checkpoints_v3`): larger validation set and test set, wider strength grid and a random-direction control.
- **Run 4** (`checkpoints_extension`): sweeps over $k \in \{1,3,5\}$, layer and strength at fixed settings.
- **Run 5** (final run): cleaned outputs, repaired history construction, per-condition tuning and a shared-basis multi-component vector (Section 8). It is reported from the notebook's summary tables (absolute means, paired comparisons against Mean Difference, the selected settings and a bar chart).

How to read the tables. Every "Δ%" is the change relative to the *no-steering* score of the same run and task. Brackets give the 95% paired-bootstrap confidence interval, and **bold** marks an interval that excludes zero. No multiple-comparison correction is applied. With several dozen comparisons, a few nominally significant results are expected by chance alone, and the discussion treats isolated significant cells accordingly. For Run 5, paired differences are in absolute score units, and Δ% is computed from the absolute means relative to that run's own unsteered baseline.

2 Method recap and evaluation protocol

2.1 StyleVector baseline

For a user u with history $P_u = \{(x_i, y_i)\}$, a general LLM produces a style-agnostic response $\hat{y}_i$ for each input x_i. Let $a^{\ell}_{p,i} = h_\ell(x_i \oplus y_i)$ and $a^{\ell}_{n,i} = h_\ell(x_i \oplus \hat{y}_i)$ be last-token hidden states at layer ℓ. The paper's style vector is the mean difference

$$\mathbf{s}^{\ell}_u = \frac{1}{|P_u|} \sum_i \left(a^{\ell}_{p,i} - a^{\ell}_{n,i}\right), \tag{1}$$

and generation is steered by $h'_\ell(x)_t = h_\ell(x)_t + \alpha\, \mathbf{s}^{\ell}_u$.

2.2 Proposed extension

Multi-component vectors. With $\Delta_i = a^{\ell}_{p,i} - a^{\ell}_{n,i}$, we take the top-$k$ principal components $\{v_1, \ldots, v_k\}$ of the symmetric set $\{\Delta_i\} \cup \{-\Delta_i\}$, i.e. the leading eigenvectors of $\sum_i \Delta_i \Delta_i^{\top}$, and combine them with weights derived from their explained variance. Run 1 rescales the combination to the norm of the mean-difference vector so that a given α is comparable between methods.

Norm-preserving steering. After the additive intervention the activation is rescaled to its original norm:

$$h'_\ell(x)_t = \|h_\ell(x)_t\| \cdot \frac{h_\ell(x)_t + p}{\|h_\ell(x)_t + p\|}, \qquad p = \alpha \mathbf{s}^{\ell}_u \text{ (or its multi-component analogue).} \tag{2}$$

The motivation is the quality collapse at large α reported in the paper's Fig. 3. The proposal therefore predicts that NP matters when the perturbation is large relative to $\|h\|$.

2.3 The 2×2 ablation and controls

The proposal's ablation grid is reproduced in Table 1. Runs 2–4 add controls that test whether any gain is genuinely *personal*:

- **Global vector** (Run 2): one non-personalised vector used for all users.
- **Shrinkage** (Run 2): the user's vector shrunk towards the global vector ($\kappa = 10$).
- **Random direction + NP** (Run 3): a random direction at the same layer and strength.

	Additive	Norm-preserving
Single vector (k=1, mean difference)	Paper's method	Ablation A
Multi-component (top-k PCA)	Ablation B	**Full proposed method**

Table 1: Planned ablation grid (as in the proposal).

3 Overview of the five runs

	Run 1 (endterm)	Run 2 (v2)	Run 3 (v3)	Run 4 (extension)	Run 5 (final)
Test users per task	300	150 (149 for LaMP-5)	300 (299 for LaMP-5)	300 (298 for LaMP-7)	300
Selection users per task	none (midterm settings reused)	40 + 40 (two validation sets)	60	none for main run; 50 for sweeps	30 disjoint tuning users (notebook default)
Max. history items	60 / 60 / 15	30 / 30 / 15	30 / 30 / 15	15 / 15 / 15	15 / 15 / 15 (notebook default)
Layers searched	16 (fixed)	12, 16, 20, 24	12, 16, 20, 24	16 (fixed); sweep 12–24	16, 20
Strength parameter	$\alpha \in \{0.25, 0.5, 0.5\}$ (fixed)	$\rho = \|p\|/\|h\| \in \{0.05, \ldots, 0.8\}$	$\rho \in \{0.05, \ldots, 1.2\}$	$\alpha \in \{0.02, 0.02, 0.5\}$; sweep $\times\{1, 4, 16, 64\}$	α tuned per condition: $\{0.25, \ldots, 1.5\}$ plain, $\{0.5, \ldots, 4\}$ NP
Multi-component vector	per-user PCA, k=3	per-user PCA, $k \in \{3, 5\}$	per-user PCA (k not recorded)	per-user PCA, $k \in \{1, 3, 5\}$	shared top-r basis (r=16), leave-one-out
Output cleaning	none (raw)	yes (raw kept)	yes (raw kept)	none (raw)	yes, all conditions
History construction	as midterm	as midterm	as midterm	as midterm	neutral tweet rewrite (LaMP-7); chat-templated, LaMP-style prompts
Steered positions	last prompt token + generated	last token	last token	last token	all prompt tokens except BOS + generated
Controls	none	global vector, shrinkage	random direction	none	per-user PCA (negative control)
Statistics	none (added here)	paired bootstrap	paired bootstrap	none (added here)	paired bootstrap vs. Mean Diff

Table 2: Design differences between runs, as recorded in the saved settings files and outputs. All runs use LLaMA-2-7B-chat as the steered model (stated in the Run 1 notebook; the later archives do not record the model identifier). Run 5 entries marked "notebook default" are the defaults of the Run 5 notebook; the selected layers and strengths are in Table 17.

Three differences matter throughout. First, Runs 2, 3 and 5 apply a text-cleaning step to generations (stripping preambles such as "Sure! Here's…" and numbering), while Runs 1 and 4 score raw generations. Second, Runs 2, 3 and 5 select hyper-parameters on held-out users, while Runs 1 and 4 use fixed settings. Third, Run 5 changes how history pairs are built and how activations are extracted (Section 8), which no earlier run did.

4 Run 1: first end-term attempt

Setup. LLaMA-2-7B-chat in 4-bit (NF4), 300 LaMP dev users per task, layer 16, fixed $\alpha = 0.25/0.5/0.5$ (LaMP-4/5/7), histories capped at $60/60/15$. Five conditions were evaluated per user: no steering, Mean Difference, Mean Difference + NP, multi-PCA (k=3) and multi-PCA + NP. The multi-component vector is a variance-weighted combination of the top three components, rescaled to the norm of the mean-difference vector. Outputs were scored raw, with no cleaning.

Task	Configuration	ROUGE-L	Δ% vs. base [95% CI]	METEOR	Δ% vs. base [95% CI]
LaMP-4 (n=300)	No steering	0.0909	–	0.1203	–
	Single vector, additive (paper)	0.0907	-0.3 [-4.6,+4.0]	0.1155	-4.0 [-9.6,+1.2]
	Single vector + NP	0.0909	-0.0 [-4.2,+4.2]	0.1162	-3.4 [-8.9,+1.6]
	Multi-PCA, additive	0.0886	-2.5 [-6.4,+1.1]	0.1143	**-5.0 [-9.5,-0.8]**
	Multi-PCA + NP (ours)	0.0895	-1.6 [-5.7,+2.3]	0.1149	**-4.6 [-9.2,-0.2]**
LaMP-5 (n=300)	No steering	0.2137	–	0.3179	–
	Single vector, additive (paper)	0.2203	+3.1 [-0.4,+6.6]	0.3131	-1.5 [-5.1,+2.0]
	Single vector + NP	0.2222	**+4.0 [+0.3,+7.8]**	0.3160	-0.6 [-4.2,+3.0]
	Multi-PCA, additive	0.2261	**+5.8 [+2.4,+9.4]**	0.3284	+3.3 [-0.2,+6.9]
	Multi-PCA + NP (ours)	0.2271	**+6.3 [+2.9,+9.8]**	0.3300	**+3.8 [+0.4,+7.3]**
LaMP-7 (n=300)	No steering	0.3584	–	0.3440	–
	Single vector, additive (paper)	0.3577	-0.2 [-4.5,+4.0]	0.3317	-3.6 [-7.9,+0.6]
	Single vector + NP	0.3554	-0.8 [-5.0,+3.4]	0.3296	-4.2 [-8.6,+0.1]
	Multi-PCA, additive	0.3579	-0.1 [-4.1,+3.7]	0.3273	**-4.8 [-9.1,-0.8]**
	Multi-PCA + NP (ours)	0.3535	-1.4 [-5.2,+2.4]	0.3210	**-6.7 [-10.9,-2.6]**

Table 3: Run 1 results. Δ% is relative to no steering. Bold: 95% interval excludes 0.

Findings.

- The paper's method produced no measurable ROUGE-L change on any task (-0.3%, $+3.1\%$, -0.2%; all intervals include zero), and METEOR point estimates were negative on LaMP-4 (-4.0%) and LaMP-7 (-3.6%) with intervals that include zero.
- LaMP-5 is the only task with significant improvements, and they are largest for the proposed method: multi-PCA + NPgives $+6.3\%$ ROUGE-L and $+3.8\%$ METEOR. Relative to the paper's method, the multi-component vector adds $+4.8\%$ METEOR (significant) and NP alone adds a small $+0.9\%$ ROUGE-L (significant).
- On LaMP-4 and LaMP-7 the multi-component variants have *lower* METEOR than no steering (-5.0% and -4.8% for multi-PCA, -4.6% and -6.7% for the full method; all significant).
- Raw generations were highly chatty: 93% (LaMP-4) and 99.7% (LaMP-5) of unsteered outputs begin with a preamble such as "Here are a few…". This inflates length and depresses ROUGE, and steering does not change it (Section 9.4). Run 1's LaMP-7 baseline is much less chatty (17%) than in Runs 2–4 (83–86%), so its LaMP-7 baseline is not comparable across runs. The archives do not record why.

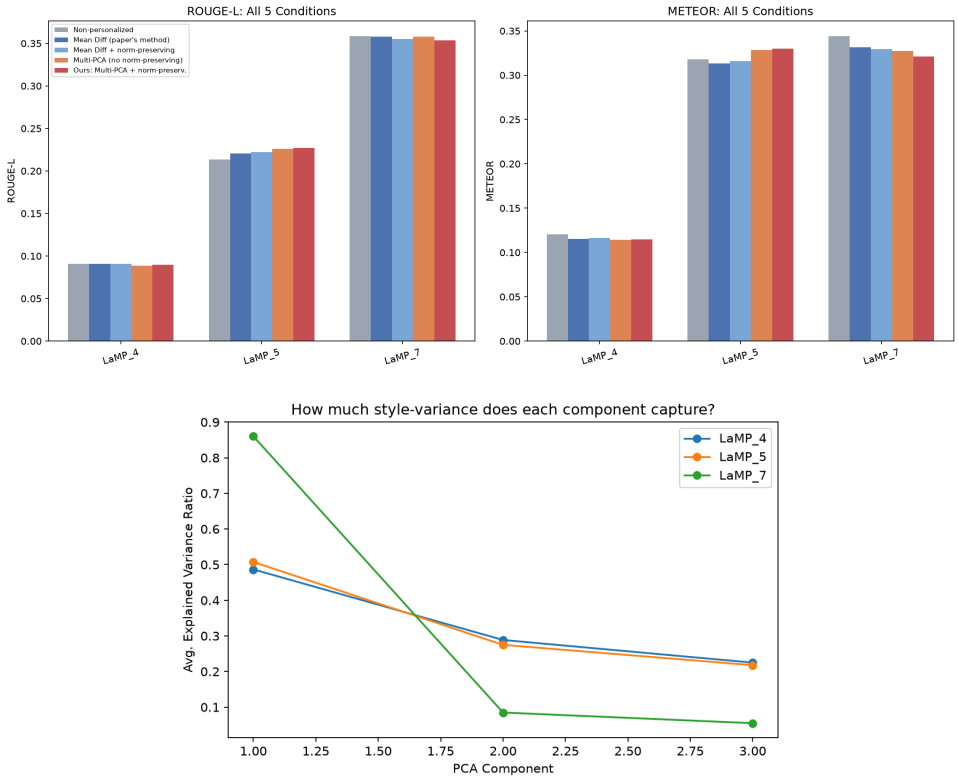


Figure 1: Run 1, original figures. Top: mean ROUGE-L and METEOR for the five conditions. Bottom: average explained-variance ratio of the PCA components.

5 Run 2: validation-tuned, cleaned, with controls

Setup. 150 test users per task and two disjoint validation sets of 40 users. The first (val1) screened the paper's method over a 4×5 grid of layers $\{12, 16, 20, 24\}$ and strengths $\rho \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$. The second (val2) evaluated the remaining families at layers $\{12, 16\}$. Selection also covered $k \in \{3, 5\}$ and $\kappa = 10$. Outputs and metrics were computed after cleaning (raw scores are stored alongside). Six families were evaluated: single additive, single + NP, global vector, shrinkage, PCA additive and PCA + NP.

Tuning outcome. In *all 18* selected configurations, ρ was the smallest value in the grid (0.05). On LaMP-5 no family beat no steering on validation (all gains between -0.0076 and -0.0122). Figure 2 shows that the validation surface is noisy and non-monotone, with isolated peaks (for example layer 16, $\rho = 0.4$ on LaMP-5) surrounded by lower values. This is the pattern expected when the signal is weak relative to the noise of 40 validation users.

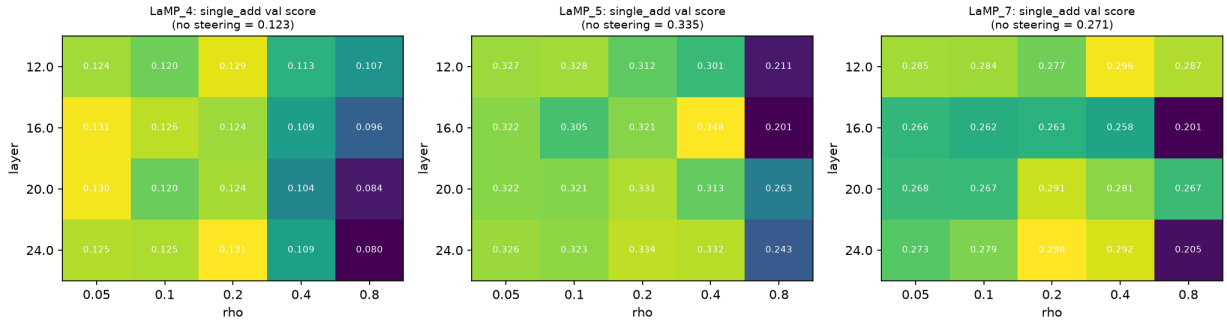


Figure 2: Run 2 validation score of the single additive method over layer and ρ (val1, n=40 per task). "No steering" is the validation baseline.

Task	Method	Layer	ρ	k	Val. score	No-steer val.	Val. gain
LaMP-4	single, add	16	0.05	1	0.1305	0.1228	+0.0077
	single + NP	16	0.05	1	0.1325	0.1228	+0.0097
	global (control)	16	0.05	1	0.1312	0.1228	+0.0084
	shrunk (control)	16	0.05	1	0.1312	0.1228	+0.0084
	PCA, add	16	0.05	3	0.1238	0.1228	+0.0010
	PCA + NP	16	0.05	3	0.1258	0.1228	+0.0030
LaMP-5	single, add	12	0.05	1	0.3268	0.3347	-0.0078
	single + NP	12	0.05	1	0.3268	0.3347	-0.0078
	global (control)	12	0.05	1	0.3225	0.3347	-0.0122
	shrunk (control)	12	0.05	1	0.3268	0.3347	-0.0078
	PCA, add	12	0.05	5	0.3270	0.3347	-0.0076
	PCA + NP	12	0.05	5	0.3270	0.3347	-0.0076
LaMP-7	single, add	12	0.05	1	0.2848	0.2707	+0.0141
	single + NP	12	0.05	1	0.2850	0.2707	+0.0144
	global (control)	12	0.05	1	0.2898	0.2707	+0.0191
	shrunk (control)	12	0.05	1	0.2865	0.2707	+0.0158
	PCA, add	12	0.05	3	0.2808	0.2707	+0.0101
	PCA + NP	12	0.05	3	0.2808	0.2707	+0.0102

Table 4: Run 2 selected settings and validation scores. "Val. gain" is the validation score minus the no-steering validation score.

Task	Configuration	ROUGE-L	Δ% vs. base [95% CI]	METEOR	Δ% vs. base [95% CI]
LaMP-4 (n=150)	No steering	0.1105	–	0.0933	–
	Single vector, additive (paper)	0.1156	+4.7 [-3.0,+14.8]	0.1006	+7.8 [-3.0,+22.8]
	Single vector + NP	0.1136	+2.8 [-3.5,+12.2]	0.1001	+7.3 [-3.0,+22.2]
	Global vector (control)	0.1097	-0.7 [-5.1,+3.8]	0.0917	-1.7 [-5.5,+2.2]
	Shrunk to global (control)	0.1123	+1.6 [-3.8,+7.9]	0.0950	+1.7 [-3.9,+8.8]
	Multi-PCA, additive	0.1136	+2.8 [-4.5,+12.6]	0.0981	+5.2 [-5.1,+19.7]
	Multi-PCA + NP (ours)	0.1127	+2.1 [-3.9,+11.2]	0.0983	+5.3 [-4.3,+19.7]
LaMP-5 (n=150)	No steering	0.3240	–	0.3269	–
	Single vector, additive (paper)	0.3229	-0.3 [-3.8,+3.0]	0.3246	-0.7 [-4.7,+2.8]
	Single vector + NP	0.3199	-1.3 [-5.1,+2.3]	0.3220	-1.5 [-5.8,+2.3]
	Global vector (control)	0.3216	-0.7 [-4.2,+2.6]	0.3228	-1.3 [-5.5,+2.6]
	Shrunk to global (control)	0.3218	-0.7 [-4.0,+2.6]	0.3221	-1.5 [-5.6,+2.3]
	Multi-PCA, additive	0.3218	-0.7 [-3.9,+2.5]	0.3252	-0.5 [-3.7,+2.7]
	Multi-PCA + NP (ours)	0.3213	-0.8 [-4.1,+2.4]	0.3235	-1.0 [-4.4,+2.3]
LaMP-7 (n=150)	No steering	0.2975	–	0.2640	–
	Single vector, additive (paper)	0.3016	+1.4 [-1.8,+4.5]	0.2746	**+4.0 [+0.4,+7.9]**
	Single vector + NP	0.2978	+0.1 [-3.1,+3.1]	0.2716	+2.9 [-0.6,+6.4]
	Global vector (control)	0.3047	+2.4 [-0.8,+5.9]	0.2796	**+5.9 [+2.3,+9.8]**
	Shrunk to global (control)	0.3033	+2.0 [-1.0,+4.9]	0.2773	**+5.0 [+1.8,+8.6]**
	Multi-PCA, additive	0.3042	+2.3 [-1.2,+5.6]	0.2754	**+4.3 [+0.7,+8.3]**
	Multi-PCA + NP (ours)	0.3035	+2.0 [-1.3,+5.3]	0.2759	**+4.5 [+0.8,+8.5]**

Table 5: Run 2 test results (n=150), cleaned outputs.

Findings.

- The paper's method beats no steering only on LaMP-7 METEOR (+4.0%, significant). Other cells are inconclusive: LaMP-4 shows large but very uncertain point estimates (+4.7% ROUGE-L, +7.8% METEOR) and LaMP-5 shows none.
- Multi-PCA and NP give no significant change relative to the single additive vector in any task or metric (Table 28, Appendix A).
- **Global-vector control.** The non-personalised global vector is statistically indistinguishable from the user vector on all tasks (Table 6). On LaMP-7 it is numerically *better* (+5.9% METEOR over no steering, significant). Shrinkage also adds nothing beyond the global vector.

Task	Comparison	ROUGE-L: Δ% of base [95% CI]	METEOR: Δ% of base [95% CI]
LaMP-4	Global vector vs. user vector	-5.3 [-14.5,+1.1]	-9.5 [-24.1,+0.7]
	Shrunk vector vs. user vector	-3.0 [-11.9,+3.1]	-6.0 [-20.0,+3.0]
	Shrunk vector vs. global vector	+2.3 [-0.4,+6.6]	**+3.5 [+0.0,+9.2]**
LaMP-5	Global vector vs. user vector	-0.4 [-2.9,+2.3]	-0.6 [-3.2,+1.9]
	Shrunk vector vs. user vector	-0.3 [-2.6,+2.3]	-0.8 [-3.3,+1.5]
	Shrunk vector vs. global vector	+0.0 [-0.8,+0.9]	-0.2 [-1.1,+0.5]
LaMP-7	Global vector vs. user vector	+1.0 [-2.3,+4.3]	+1.9 [-1.4,+5.2]
	Shrunk vector vs. user vector	+0.6 [-2.4,+3.3]	+1.0 [-1.8,+3.8]
	Shrunk vector vs. global vector	-0.5 [-2.5,+1.0]	-0.9 [-3.0,+0.6]

Table 6: Run 2 controls: user vector vs. global and shrunk vectors (paired, as % of the no-steering score).

How personal are the user vectors? The saved diagnostic compares each user's vector to the global vector (Table 7, Figure 3). On LaMP-7, 95% of users have cosine similarity above 0.9 with the global vector, so the user vector is nearly the global one. LaMP-4 is bimodal: 57% of users are nearly identical to the global vector and 43% are nearly orthogonal to it, with much smaller norm. In addition, only 102 of the 150 LaMP-4 test users have distinct (cosine, norm-ratio, history-length) triples, which suggests duplicated profiles in the dev split. LaMP-5 has the most individual vectors (mean cosine 0.65, none above 0.9) but 4.7% of its users have a vector norm more than 5× the global norm, and steering had no effect there.

Task	Users	Mean cos	Median cos	Users with cos > 0.9 (%)	Median $\|s_u\|/\|s_{\text{glob}}\|$
LaMP-4	150	0.562	0.995	56.7	0.57
LaMP-5	149	0.652	0.666	0.0	1.04
LaMP-7	150	0.950	0.957	95.3	1.05

Table 7: Run 2 personalisation diagnostic: similarity between each test user's mean-difference vector and the global mean-difference vector.

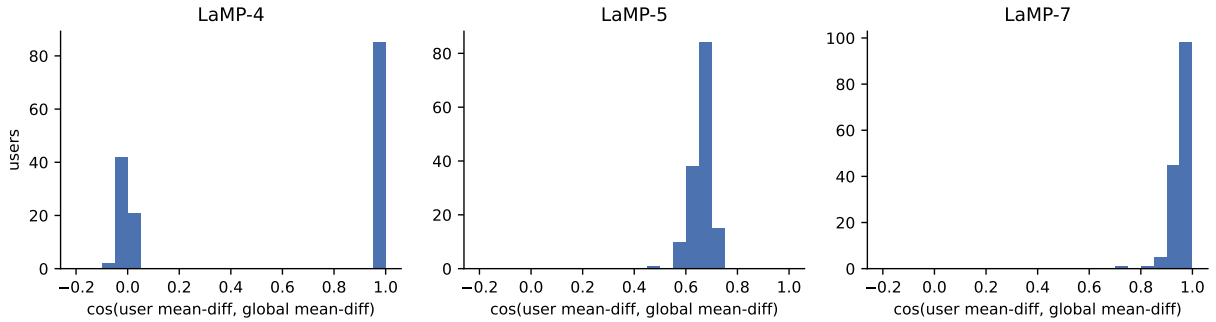


Figure 3: Distribution of cosine similarity between user and global style vectors (Run 2).

6 Run 3: larger validation and test sets, random control

Setup. 300 test users per task, 60 validation users, layers $\{12, 16, 20, 24\}$ and a wider strength grid $\rho \in \{0.05, \ldots, 0.8, 1.2\}$. Only the proposed family (PCA additive and PCA + NP) is evaluated, together with a random-direction control and no steering.

Task	Method	Layer	ρ	Val. gain	95% half-width	Val. users
LaMP-4	PCA, add	20	0.05	+0.0056	0.0163	60
	PCA + NP	20	0.05	+0.0082	0.0158	60
LaMP-5	PCA, add	20	0.05	-0.0043	0.0094	60
	PCA + NP	20	0.05	-0.0052	0.0095	60
LaMP-7	PCA, add	12	0.05	+0.0077	0.0112	60
	PCA + NP	12	0.05	+0.0087	0.0114	60

Table 8: Run 3 selected settings. The 95% half-width is that of the validation gain.

As in Run 2, the selected strength is the grid minimum ($\rho = 0.05$) for all six selections. The validation gains ($\leq +0.0087$) are smaller than their half-widths (0.009–0.016), so no selected configuration is distinguishable from no steering on validation. Figure 4 shows the full surface. Quality is roughly flat for $\rho \leq 0.2$, degrades by $\rho = 0.8$ and collapses at $\rho = 1.2$, most strongly on LaMP-5 (gain ≈ -0.3). NP and additive steering behave almost identically throughout; the largest difference is at $\rho = 1.2$ on LaMP-7, where NP is less bad (-0.10 vs. -0.14).

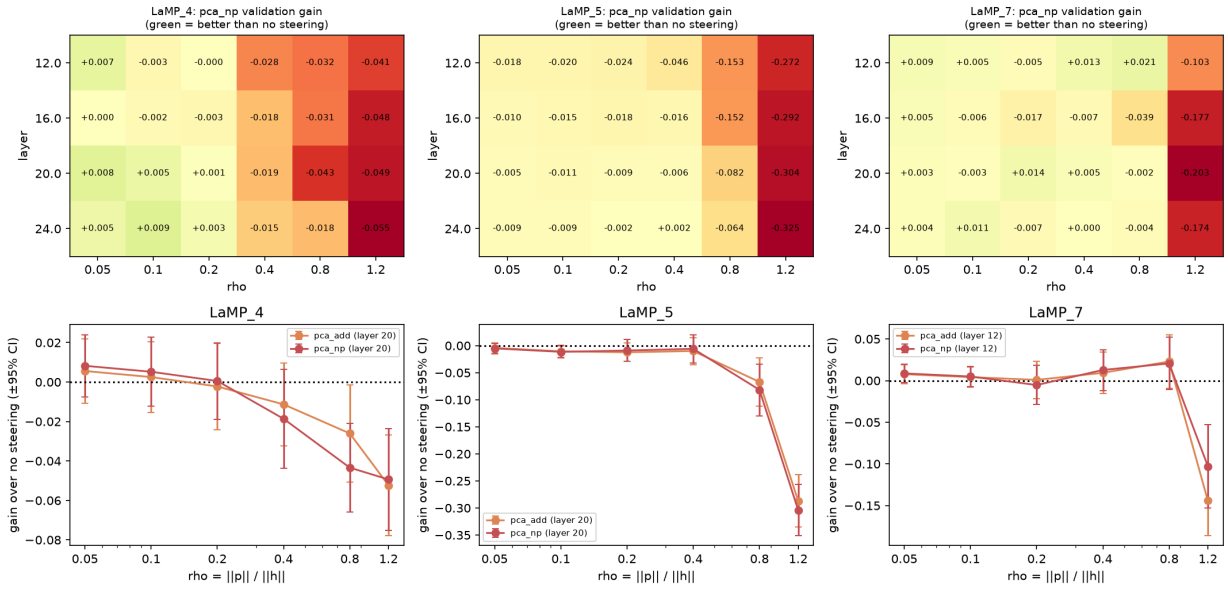


Figure 4: Run 3 validation gain of PCA + NPover no steering (top, per layer and ρ) and gain with 95% intervals at the selected layer (bottom).

Task	Configuration	ROUGE-L	Δ% vs. base [95% CI]	METEOR	Δ% vs. base [95% CI]
LaMP-4 (n=300)	No steering	0.1098	–	0.0966	–
	Multi-PCA, additive	0.1132	+3.1 [-2.0,+8.8]	0.1027	+6.3 [-0.1,+14.2]
	Multi-PCA + NP (ours)	0.1123	+2.3 [-1.9,+7.6]	0.1020	+5.6 [-0.3,+13.3]
	Random direction + NP (control)	0.1092	-0.6 [-5.8,+5.2]	0.0964	-0.2 [-6.6,+7.8]
LaMP-5 (n=300)	No steering	0.3196	–	0.3348	–
	Multi-PCA, additive	0.3205	+0.3 [-1.1,+1.7]	0.3346	-0.1 [-2.3,+2.2]
	Multi-PCA + NP (ours)	0.3197	+0.0 [-1.4,+1.5]	0.3340	-0.2 [-2.7,+2.2]
	Random direction + NP (control)	0.3209	+0.4 [-0.9,+2.0]	0.3344	-0.1 [-1.9,+1.7]
LaMP-7 (n=300)	No steering	0.2938	–	0.2618	–
	Multi-PCA, additive	0.2975	+1.3 [-0.7,+3.2]	0.2668	+1.9 [-0.2,+4.1]
	Multi-PCA + NP (ours)	0.2988	+1.7 [-0.3,+3.6]	0.2678	**+2.3 [+0.1,+4.6]**
	Random direction + NP (control)	0.2944	+0.2 [-1.5,+1.9]	0.2648	+1.2 [-0.4,+2.9]

Table 9: Run 3 test results, cleaned outputs.

Findings.

- Vs. no steering, the proposed method improves only LaMP-7 METEOR significantly ($+2.3\%$, lower bound close to zero). LaMP-4 METEOR is borderline ($+5.6\%$ to $+6.3\%$ with a lower bound at or just below zero) and LaMP-5 is flat.
- **Random-direction control.** A random direction with the same strength gives -0.6%, $+0.4\%$ and $+0.2\%$ ROUGE-L. PCA + NP is not significantly better than the random direction on any task or metric in our recomputation (Table 10). One case is borderline (LaMP-4 METEOR, $+5.8\%$, lower bound at zero), which the Run 3 summary file flags as significant under its own bootstrap.
- NP on top of PCA is inconclusive, with one small significant *decrease* (LaMP-5 ROUGE-L, -0.25%).

Task	Comparison	ROUGE-L: Δ% of base [95% CI]	METEOR: Δ% of base [95% CI]
LaMP-4	PCA+NP vs. random direction+NP	+2.9 [-2.0,+7.8]	+5.8 [-0.0,+11.6]
	Random direction+NP vs. no steering	-0.6 [-5.8,+5.2]	-0.2 [-6.4,+7.6]
LaMP-5	PCA+NP vs. random direction+NP	-0.4 [-2.4,+1.5]	-0.1 [-2.9,+2.6]
	Random direction+NP vs. no steering	+0.4 [-1.0,+2.0]	-0.1 [-1.9,+1.7]
LaMP-7	PCA+NP vs. random direction+NP	+1.5 [-0.5,+3.5]	+1.1 [-1.1,+3.5]
	Random direction+NP vs. no steering	+0.2 [-1.5,+1.9]	+1.2 [-0.5,+2.9]

Table 10: Run 3 random-direction control (paired, as % of the no-steering score).

7 Run 4: k, layer and strength sweeps

Setup. Fixed layer 16 with $\alpha = 0.02/0.02/0.5$ for LaMP-4/5/7, 300 test users, evaluating single and PCA vectors with $k \in \{1,3,5\}$, each additive and NP. Three sweeps use 50 users per task: layers $\{12,16,20,24\}$, strength multipliers $\{1,4,16,64\}$ of the base α, and k. Outputs and metrics were *not* cleaned.

Task	Configuration	ROUGE-L	Δ% vs. base [95% CI]	METEOR	Δ% vs. base [95% CI]
LaMP-4 (n=300)	No steering	0.0971	–	0.1206	–
	Single vector, additive (paper)	0.0963	**-0.8 [-1.7,-0.1]**	0.1207	+0.0 [-1.2,+1.1]
	Single vector + NP	0.0963	**-0.8 [-1.7,-0.1]**	0.1202	-0.4 [-2.0,+1.0]
	PCA k=1, additive	0.0964	-0.8 [-1.7,+0.1]	0.1201	-0.5 [-2.2,+1.2]
	PCA k=1 + NP	0.0965	-0.6 [-1.6,+0.2]	0.1206	-0.0 [-1.8,+1.6]
	Multi-PCA, additive	0.0961	-1.0 [-2.4,+0.2]	0.1200	-0.5 [-1.9,+0.8]
	Multi-PCA + NP (ours)	0.0959	**-1.2 [-2.5,-0.0]**	0.1198	-0.7 [-2.1,+0.5]
	PCA k=5, additive	0.0961	-1.0 [-2.3,+0.1]	0.1197	-0.8 [-1.9,+0.1]
	PCA k=5 + NP	0.0965	-0.6 [-2.1,+0.7]	0.1202	-0.4 [-1.7,+0.9]
LaMP-5 (n=300)	No steering	0.2118	–	0.3122	–
	Single vector, additive (paper)	0.2106	-0.5 [-1.5,+0.3]	0.3116	-0.2 [-1.5,+0.8]
	Single vector + NP	0.2104	-0.6 [-1.6,+0.2]	0.3116	-0.2 [-1.5,+0.8]
	PCA k=1, additive	0.2106	-0.6 [-1.6,+0.3]	0.3115	-0.2 [-1.5,+0.8]
	PCA k=1 + NP	0.2104	-0.7 [-1.6,+0.2]	0.3113	-0.3 [-1.6,+0.8]
	Multi-PCA, additive	0.2089	**-1.3 [-2.8,-0.2]**	0.3095	-0.9 [-2.3,+0.2]
	Multi-PCA + NP (ours)	0.2095	-1.1 [-2.6,+0.1]	0.3107	-0.5 [-2.0,+0.8]
	PCA k=5, additive	0.2086	**-1.5 [-3.1,-0.3]**	0.3094	-0.9 [-2.4,+0.2]
	PCA k=5 + NP	0.2086	**-1.5 [-3.0,-0.3]**	0.3093	-0.9 [-2.4,+0.2]
LaMP-7 (n=300)	No steering	0.2429	–	0.2644	–
	Single vector, additive (paper)	0.3133	**+29.0 [+23.6,+34.5]**	0.3100	**+17.2 [+11.8,+22.6]**
	Single vector + NP	0.3041	**+25.2 [+19.9,+30.5]**	0.2977	**+12.6 [+7.3,+17.8]**
	PCA k=1, additive	0.3121	**+28.5 [+23.2,+33.9]**	0.3085	**+16.7 [+11.3,+22.0]**
	PCA k=1 + NP	0.3039	**+25.1 [+19.8,+30.5]**	0.2988	**+13.0 [+7.7,+18.2]**
	Multi-PCA, additive	0.3132	**+29.0 [+23.6,+34.5]**	0.3086	**+16.7 [+11.1,+22.6]**
	Multi-PCA + NP (ours)	0.3081	**+26.9 [+21.2,+32.6]**	0.3051	**+15.4 [+9.7,+21.0]**
	PCA k=5, additive	0.3166	**+30.4 [+24.9,+35.9]**	0.3146	**+19.0 [+13.5,+24.6]**
	PCA k=5 + NP	0.3082	**+26.9 [+21.2,+32.7]**	0.3084	**+16.7 [+10.9,+22.5]**

Table 11: Run 4 main results (raw outputs, n=300).

Findings: LaMP-4 and LaMP-5. With $\alpha = 0.02$ the perturbation is minimal: about 90% of the stored steered outputs are identical to the unsteered ones. The few significant differences are small and negative (for example -0.8% ROUGE-L for the single vector on LaMP-4, and -1.3% for k=3 and -1.5% for k=5 on LaMP-5). Increasing the number of components does not help: on LaMP-5, k=5 is significantly worse than k=1 on ROUGE-L (-0.9%) and k=3 is significantly worse than k=1 on METEOR (-0.6%), and no comparison between values of k is significantly positive (Table 12).

Findings: LaMP-7. Here steering changes almost every output (98% differ from the unsteered one) and every variant improves massively over no steering: $+29\%$ ROUGE-L for the single vector. The gain is not specific to the proposed method (k=1: $+28.5\%$, k=3: $+29.0\%$, k=5: $+30.4\%$), and k has no significant effect. NP is significantly *worse* than the additive version here (-3.8% ROUGE-L for the single vector, -3.5% for k=5). The size of this effect prompted a check of the outputs, reported in the next subsection.

Task	Comparison	ROUGE-L: Δ% of base [95% CI]	METEOR: Δ% of base [95% CI]
LaMP-4	PCA k=1 vs. single (mean-diff)	+0.0 [-0.5,+0.6]	-0.5 [-1.9,+0.8]
	PCA k=3 vs. k=1 (additive)	-0.3 [-1.5,+0.7]	-0.0 [-1.7,+1.7]
	PCA k=5 vs. k=1 (additive)	-0.2 [-1.6,+0.9]	-0.3 [-2.1,+1.5]
	PCA k=5 vs. single (additive)	-0.2 [-1.5,+0.8]	-0.8 [-2.1,+0.4]
	NP on top of PCA k=5	+0.3 [-0.0,+1.1]	+0.4 [+0.0,+1.2]
LaMP-5	PCA k=1 vs. single (mean-diff)	-0.0 [-0.3,+0.2]	-0.0 [-0.2,+0.1]
	PCA k=3 vs. k=1 (additive)	-0.8 [-2.1,+0.1]	**-0.6 [-1.5,-0.0]**
	PCA k=5 vs. k=1 (additive)	**-0.9 [-2.3,-0.0]**	-0.7 [-1.7,+0.1]
	PCA k=5 vs. single (additive)	**-1.0 [-2.3,-0.1]**	-0.7 [-1.7,+0.1]
	NP on top of PCA k=5	+0.0 [-0.0,+0.0]	-0.0 [-0.0,+0.0]
LaMP-7	PCA k=1 vs. single (mean-diff)	-0.5 [-1.7,+0.4]	-0.6 [-1.6,+0.2]
	PCA k=3 vs. k=1 (additive)	+0.5 [-3.8,+4.8]	+0.1 [-4.1,+4.3]
	PCA k=5 vs. k=1 (additive)	+1.9 [-1.9,+5.8]	+2.3 [-1.5,+6.3]
	PCA k=5 vs. single (additive)	+1.3 [-2.3,+5.3]	+1.8 [-2.1,+5.8]
	NP on top of PCA k=5	**-3.5 [-6.0,-1.1]**	**-2.3 [-4.6,-0.1]**

Table 12: Run 4 effect of the number of components (paired, as % of the no-steering score).

7.1 Diagnosing the LaMP-7 gain

Of the 298 LaMP-7 users, 85% had an unsteered output that begins with "Sure! Here's a paraphrase…". Steering reduces this to 21% (Table 25), and Table 14 shows examples. Because raw outputs were scored, much of the "gain" is simply the removal of a preamble. To separate the format effect from the rest, all Run 4 outputs were rescored with ROUGE-L after applying the same cleaning rule as Runs 2 and 3 (Table 13; METEOR was not rescored).

Task	No steering	Single, add	Single vs. none (%)	Multi + NP vs. none (%)	Multi + NP vs. single (%)
LaMP-4 (n=300)	0.1144	0.1140	-0.3 [-1.1,+0.4]	-1.0 [-2.7,+0.2]	-0.7 [-2.2,+0.3]
LaMP-5 (n=300)	0.3209	0.3195	-0.4 [-1.5,+0.5]	**-1.2 [-2.6,-0.0]**	**-0.7 [-1.9,-0.0]**
LaMP-7 (n=300)	0.2910	0.3273	**+12.5 [+7.8,+17.1]**	**+9.9 [+5.0,+14.6]**	-2.6 [-6.5,+1.3]

Table 13: Run 4 rescored with output cleaning (ROUGE-L only). Single and multi-PCA + NP (k=3).

After cleaning, the LaMP-7 gain of the single vector falls from $+29\%$ to $+12.5\%$ (significant), and LaMP-4 and LaMP-5 are unchanged or slightly negative (multi-PCA + NPon LaMP-5: -1.2%, interval touching zero). So more than half of the apparent gain was formatting, and a significant effect remains on LaMP-7. The multi-component NP variant ($+9.9\%$) is not better than the single additive vector (difference -2.6%, interval includes zero). Within the 253 users whose unsteered output was chatty, the cleaned ROUGE-L gain is $+0.038$ ($[+0.023, +0.053]$). Within the 45 users whose output was not chatty it is $+0.026$ with an interval ($[-0.006, +0.057]$) that includes zero, so the effect is not confined to the format fix. Whether the remaining gain is *personal* cannot be decided from Run 4 alone, because it has no global-vector control. Run 2's control on the same task and the Run 2 diagnostic (95% of LaMP-7 user vectors nearly equal the global vector) suggest it is mostly a shared "tweet register" shift rather than user-specific style.

Gold tweet	@Funkybabygirl Hopelessly Devoted to you from grease Lool Xx
No steering	Sure! Here's a paraphrase of the tweet: @Funkybabygirl found the quote funny and expressed this by adding "lool"
Single, add	@Funkybabygirl found it funny and expressed this by adding "lool" and two kisses after quoting the song "Hopelessly Devoted to You
Multi + NP	@Funkybabygirl found it funny and expressed this by adding "lool" and two kisses after quoting the song "Hopelessly Devoted to You
Gold tweet	@markotu sounds good to me! you go out loads!!! lol x jealous! although i have coffee and french pastries tomorrow, no where near as good
No steering	Sure! Here's a paraphrase of the tweet: "I concur with you, @markotu. It appears that you frequently go out and I'
Single, add	"I see you, @markotu, always going out and about. My plans for tomorrow are coffee and French pastries, but it doesn't compare to your adventures
Multi + NP	"I see you're always going out and about, @markotu. I'm feeling a bit jealous. Tomorrow I have coffee and pastries, but it's

Table 14: Qualitative LaMP-7 examples from Run 4 (stored predictions are truncated at generation length; emoji removed).

7.2 Layer and strength sweeps

Task	Metric	No steering	L=12	L=16	L=20	L=24
LaMP-4 (n=50)	ROUGE-L: Single, add	0.1012	0.0992	0.1012	0.1011	0.1002
	ROUGE-L: PCA k=3 + NP	0.1012	0.0993	0.1012	0.1012	0.1010
	METEOR: Single, add	0.1281	0.1273	0.1281	0.1244	0.1243
	METEOR: PCA k=3 + NP	0.1281	0.1267	0.1274	0.1237	0.1237
LaMP-5 (n=50)	ROUGE-L: Single, add	0.2231	0.2243	0.2229	0.2228	0.2227
	ROUGE-L: PCA k=3 + NP	0.2231	0.2231	0.2227	0.2238	0.2239
	METEOR: Single, add	0.3270	0.3270	0.3276	0.3267	0.3266
	METEOR: PCA k=3 + NP	0.3270	0.3270	0.3254	0.3282	0.3275
LaMP-7 (n=50)	ROUGE-L: Single, add	0.2448	0.2491	0.3025	0.3138	0.3082
	ROUGE-L: PCA k=3 + NP	0.2448	0.2518	0.2777	0.2938	0.2883
	METEOR: Single, add	0.2664	0.2820	0.2915	0.3154	0.3152
	METEOR: PCA k=3 + NP	0.2664	0.2850	0.2806	0.2843	0.2915

Table 15: Run 4 layer sweep (n=50 per task), raw outputs.

Layer matters only where there is an effect: on LaMP-7 the single additive vector improves over no steering at layers 16, 20 and 24 (intervals exclude zero at all three) but not at layer 12, with multi-PCA + NP scoring below the single additive vector at each of these layers (differences not significant). On LaMP-4 and LaMP-5 no layer changes either metric by more than 0.005 relative to no steering.

Task	Method	No steering	×1	×4	×16	×64
LaMP-4 (n=50)	ROUGE-L: Single, add	0.1012	0.1012	0.0969	0.0977	0.0775
	ROUGE-L: Single + NP	0.1012	0.1012	0.0969	0.0978	0.0833
	ROUGE-L: PCA k=3, add	0.1012	0.0987	0.1001	0.0984	0.0874
	ROUGE-L: PCA k=3 + NP	0.1012	0.1012	0.1014	0.0992	0.0944
	METEOR: Single, add	0.1281	0.1281	0.1226	0.1164	0.0883
	METEOR: Single + NP	0.1281	0.1238	0.1226	0.1156	0.0982
	METEOR: PCA k=3, add	0.1281	0.1253	0.1297	0.1218	0.1014
	METEOR: PCA k=3 + NP	0.1281	0.1274	0.1306	0.1263	0.1038
LaMP-5 (n=50)	ROUGE-L: Single, add	0.2231	0.2229	0.2134	0.2155	0.1775
	ROUGE-L: Single + NP	0.2231	0.2240	0.2134	0.2130	0.1666
	ROUGE-L: PCA k=3, add	0.2231	0.2227	0.2138	0.2134	0.2003
	ROUGE-L: PCA k=3 + NP	0.2231	0.2227	0.2138	0.2119	0.1900
	METEOR: Single, add	0.3270	0.3276	0.3097	0.3064	0.2323
	METEOR: Single + NP	0.3270	0.3276	0.3097	0.3046	0.1973
	METEOR: PCA k=3, add	0.3270	0.3254	0.3098	0.3084	0.2578
	METEOR: PCA k=3 + NP	0.3270	0.3254	0.3090	0.3084	0.2500
LaMP-7 (n=50)	ROUGE-L: Single, add	0.2448	0.3025	0.0361	0.0000	0.0000
	ROUGE-L: Single + NP	0.2448	0.2880	0.0262	0.0064	0.0091
	ROUGE-L: PCA k=3, add	0.2448	0.2974	0.0303	0.0000	0.0000
	ROUGE-L: PCA k=3 + NP	0.2448	0.2777	0.0364	0.0095	0.0147
	METEOR: Single, add	0.2664	0.2915	0.0281	0.0086	0.0012
	METEOR: Single + NP	0.2664	0.2751	0.0194	0.0071	0.0075
	METEOR: PCA k=3, add	0.2664	0.2912	0.0260	0.0088	0.0000
	METEOR: PCA k=3 + NP	0.2664	0.2806	0.0261	0.0112	0.0098

Table 16: Run 4 strength sweep (n=50 per task): multiples of the base α, raw outputs.

The strength sweep (Table 16, Figure 5) is the experiment closest to the paper's Fig. 3:

- **Collapse.** On LaMP-7 a 4× multiplier ($\alpha = 2$) already destroys the output (ROUGE-L ≈ 0.03 for all four methods, with roughly 94–98% degenerate outputs in the saved plot). NP does *not* prevent this.
- **Graceful degradation.** At 64× on LaMP-4 and LaMP-5 the multi-component vector degrades less than the single vector: ROUGE-L is 0.087 (additive) and 0.094 (NP) for k=3 against 0.078 and 0.083 for the single vector on LaMP-4, and 0.200 and 0.190 against 0.178 and 0.167 on LaMP-5. This ordering is consistent across both tasks, but the paired differences between multi-PCA and the single vector at 64× are not significant at n=50 (for example, LaMP-4 additive $+0.010$, 95% interval $[-0.006, +0.026]$).
- **NP at extreme strength.** NP helps on LaMP-4 (single: $+0.006$, multi: $+0.007$) and hurts on LaMP-5 (single: -0.011, multi: -0.010), with all intervals including zero. At 64× the single additive vector is significantly below no steering on both tasks, whereas multi-PCA + NPis significantly below only on LaMP-5.

This is the only evidence in Runs 1–4 that is directionally consistent with the proposal's stability argument. It is a descriptive result on 50 users and cannot be stated as established.

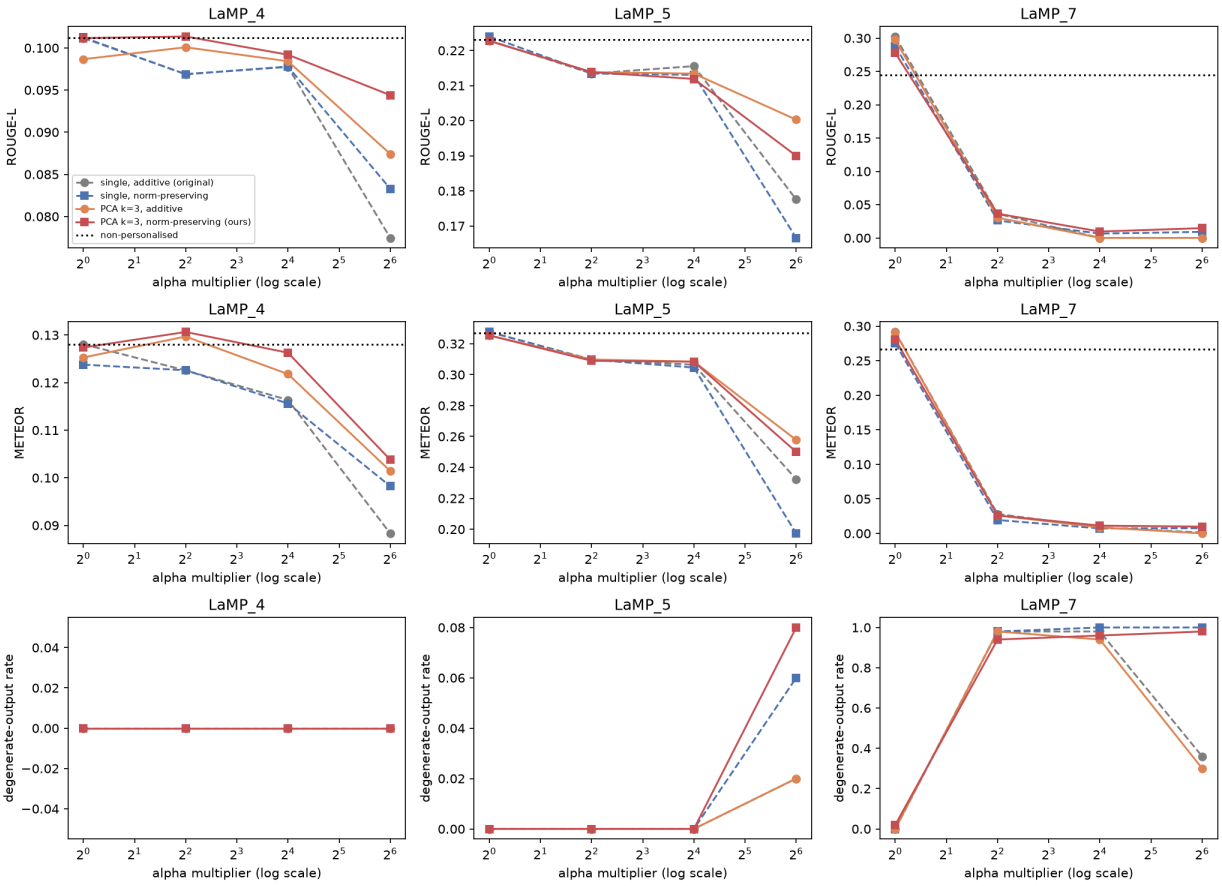


Figure 5: Run 4 strength sweep (original figure): ROUGE-L, METEOR and degenerate-output rate vs. multiplier.

8 Run 5: corrected pipeline (final run)

8.1 What changed and what was available

The Run 5 notebook was written after the analysis of Runs 1–4 to remove the problems identified there. Relative to Run 1 it differs in the following ways, all introduced at once:

1. **Cleaned outputs** (stripping "Sure! Here's…", quotes and extra lines) for the style-agnostic responses, the unsteered baseline and every steered condition.

2. **LaMP-7 history inputs.** Run 1 used the constant input "Write a short tweet." for every tweet, so all style-agnostic responses were nearly identical. Run 5 first rewrites each tweet neutrally and uses that as the input.

3. **Activations match inference.** Last-token activations are taken from the chat-templated prompt followed by the response, using LaMP-style prompt wording, so that extraction and steering see the same context.

4. **Steering positions.** All prompt tokens except BOS plus every generated token (Run 1: last prompt token plus generated tokens).

5. **Per-condition tuning** of layer ($\{16, 20\}$) and α on tuning users disjoint from the evaluation users: $\alpha \in \{0.25, 0.5, 1, 1.5\}$ for additive and $\{0.5, 1, 2, 4\}$ for NP conditions, so that NP is evaluated at larger strengths.

6. **Shared-basis multi-component vector.** Instead of per-user PCA, a top-r basis ($r = 16$) is learned from all users' mean deltas (history only, no gold outputs, leave-one-out), and each user's mean difference is projected onto it. The motivation is that a per-user estimate from 15–60 items in 4096

dimensions is too noisy for per-user PCA to help, while a basis estimated from hundreds of users is better conditioned.

7. **Six conditions:** no steering, Mean Difference (paper), Mean Difference + NP, per-user PCA k=3 (the Run 1 design, kept as a control), shared basis, and shared basis + NP(the full proposed method).

Evaluation uses the same sampling procedure as Run 1 (300 users per task from the shuffled pool). The results below are taken from the notebook's summary output: absolute means, paired bootstrap comparisons against Mean Difference, the selected settings and a bar chart. Improvements over no steering are computed from the absolute means, so confidence intervals are reported for the comparisons between steered conditions (Table 19).

8.2 Selected settings

Task	Configuration	Layer	α	Tuning score
LaMP-4	Mean Diff (paper)	16	1.5	0.1340
	Mean Diff + NP	16	1	0.0854
	PCA-k (per-user, k=3)	16	1.5	0.1372
	Multi (shared basis)	16	1.5	0.1355
	Multi (shared basis) + NP (ours)	16	1	0.0831
LaMP-5	Mean Diff (paper)	20	1	0.4009
	Mean Diff + NP	20	1	0.3870
	PCA-k (per-user, k=3)	16	0.25	0.3976
	Multi (shared basis)	16	1	0.3892
	Multi (shared basis) + NP (ours)	20	1	0.3711
LaMP-7	Mean Diff (paper)	20	0.25	0.4238
	Mean Diff + NP	16	0.5	0.3355
	PCA-k (per-user, k=3)	20	0.25	0.4275
	Multi (shared basis)	20	0.25	0.4071
	Multi (shared basis) + NP (ours)	16	0.5	0.3273

Table 17: Run 5 settings selected per condition on the tuning users. "Tuning score" is the mean of ROUGE-L and METEOR on the tuning users.

The grids ($\{16, 20\}$ layers; $\alpha \in \{0.25, 0.5, 1, 1.5\}$ additive, $\{0.5, 1, 2, 4\}$ NP) match the notebook defaults. Three observations follow from the table. (i) Additive conditions chose layer 16 on LaMP-4 and layer 20 on LaMP-5 and LaMP-7, with $\alpha = 1.5$ on LaMP-4 (the largest value in the additive grid), 1.0 on LaMP-5 and 0.25 on LaMP-7. (ii) Although NP conditions were offered strengths up to 4, none selected more than 1.0 (1.0, 1.0 and 0.5), so larger strengths were not preferred even with norm preservation. (iii) On the tuning users themselves, the NP conditions score below their additive counterparts on all three tasks (for example LaMP-4: 0.085 and 0.083 against 0.134 and 0.136; LaMP-7: 0.335 and 0.327 against 0.424 and 0.407), so the NP deficit seen on the evaluation users is already visible on independent users.

8.3 Results

Task	Configuration	ROUGE-L	Δ% vs. no steering	METEOR	Δ% vs. no steering
LaMP-4 (n=300)	No steering	0.0569	–	0.0574	–
	Mean Diff (paper)	0.0609	+7.0	0.1436	+150.2
	Mean Diff + NP	0.0676	+18.8	0.0844	+47.0
	PCA-k (per-user, k=3)	0.0636	+11.8	0.1590	+177.0
	Multi (shared basis)	0.0553	-2.8	0.1341	+133.6
	Multi (shared basis) + NP (ours)	0.0679	+19.3	0.0905	+57.7
	Paper (StyleVector)	–	+3.2	–	+2.5
LaMP-5 (n=300)	No steering	0.2578	–	0.3076	–
	Mean Diff (paper)	0.3141	+21.8	0.3411	+10.9
	Mean Diff + NP	0.3100	+20.2	0.3425	+11.3
	PCA-k (per-user, k=3)	0.3178	+23.3	0.3392	+10.3
	Multi (shared basis)	0.3019	+17.1	0.3321	+8.0
	Multi (shared basis) + NP (ours)	0.3036	+17.8	0.3345	+8.7
	Paper (StyleVector)	–	+25.8	–	+10.2
LaMP-7 (n=300)	No steering	0.3001	–	0.4055	–
	Mean Diff (paper)	0.3527	+17.5	0.4147	+2.3
	Mean Diff + NP	0.2989	-0.4	0.3620	-10.7
	PCA-k (per-user, k=3)	0.3522	+17.4	0.4148	+2.3
	Multi (shared basis)	0.3512	+17.0	0.4121	+1.6
	Multi (shared basis) + NP (ours)	0.2931	-2.3	0.3665	-9.6
	Paper (StyleVector)	–	+12.8	–	+17.5

Table 18: Run 5 results (n=300). Δ% is relative to the unsteered baseline of the same run. The paper's relative improvements are shown for reference.

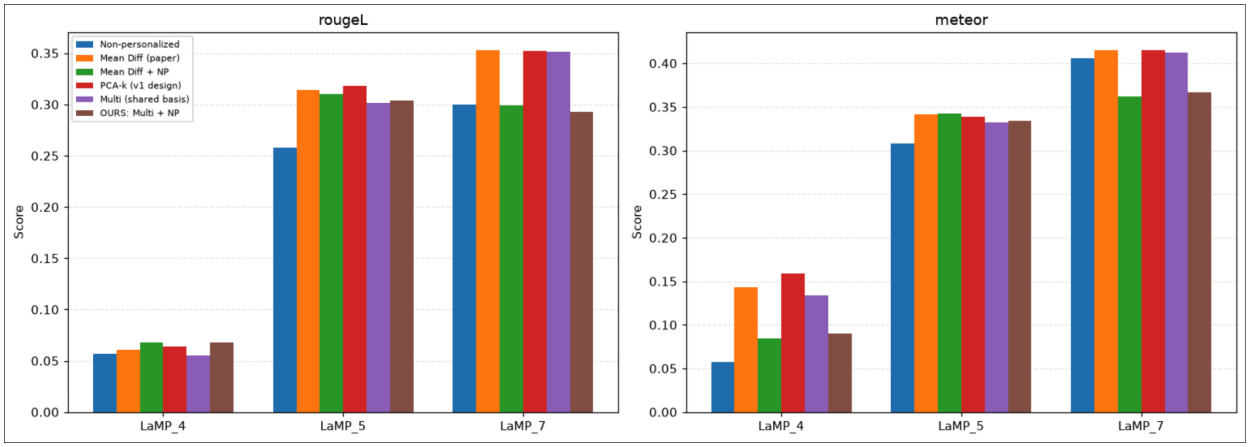


Figure 6: Run 5 mean ROUGE-L and METEOR per condition (original notebook figure, absolute-table values).

The paper's method improves ROUGE-L on all three tasks (+7.0%, +21.8%, +17.5% against the paper's +3.2%, +25.8%, +12.8%) and METEOR on LaMP-5 (+10.9% against +10.2%). On LaMP-4 the METEOR gain is large in relative terms (+150%) because the unsteered METEOR is low (0.057); the steered score is 0.144. On LaMP-7 the METEOR gain is +2.3% against the paper's +17.5%. The three additive steered conditions (Mean Diff, per-user PCA, shared basis) differ from one another by at most 0.016 in ROUGE-L and 0.025 in METEOR on any task. The two NP conditions behave differently: on LaMP-7 their ROUGE-L is at the unsteered level (0.299 and 0.293 against 0.300) and their METEOR is below it, and on LaMP-4 their METEOR is lower than that of the additive conditions.

8.4 Paired comparisons against Mean Difference

Task	Configuration − Mean Diff	ROUGE-L: diff [95% CI]	METEOR: diff [95% CI]
LaMP-4	Mean Diff + NP	+0.0067 [-0.0017,+0.0147]	**-0.0591 [-0.0974,-0.0259]**
	Multi (shared basis)	-0.0056 [-0.0136,+0.0017]	-0.0094 [-0.0196,+0.0001]
	Multi (shared basis) + NP (ours)	+0.0070 [-0.0026,+0.0162]	**-0.0531 [-0.0920,-0.0186]**
	PCA-k (per-user, k=3)	+0.0026 [-0.0045,+0.0096]	**+0.0155 [+0.0015,+0.0315]**
LaMP-5	Mean Diff + NP	-0.0042 [-0.0108,+0.0020]	+0.0014 [-0.0056,+0.0090]
	Multi (shared basis)	-0.0123 [-0.0258,+0.0013]	-0.0090 [-0.0220,+0.0042]
	Multi (shared basis) + NP (ours)	-0.0105 [-0.0232,+0.0015]	-0.0066 [-0.0186,+0.0051]
	PCA-k (per-user, k=3)	+0.0037 [-0.0093,+0.0164]	-0.0019 [-0.0148,+0.0109]
LaMP-7	Mean Diff + NP	**-0.0538 [-0.0718,-0.0365]**	**-0.0526 [-0.0722,-0.0326]**
	Multi (shared basis)	-0.0015 [-0.0068,+0.0035]	-0.0026 [-0.0085,+0.0030]
	Multi (shared basis) + NP (ours)	**-0.0596 [-0.0772,-0.0428]**	**-0.0482 [-0.0709,-0.0248]**
	PCA-k (per-user, k=3)	-0.0005 [-0.0061,+0.0047]	+0.0001 [-0.0055,+0.0060]

Table 19: Run 5 paired bootstrap differences against the paper's Mean Difference method, in score units (from the notebook output). Bold: 95% interval excludes zero.

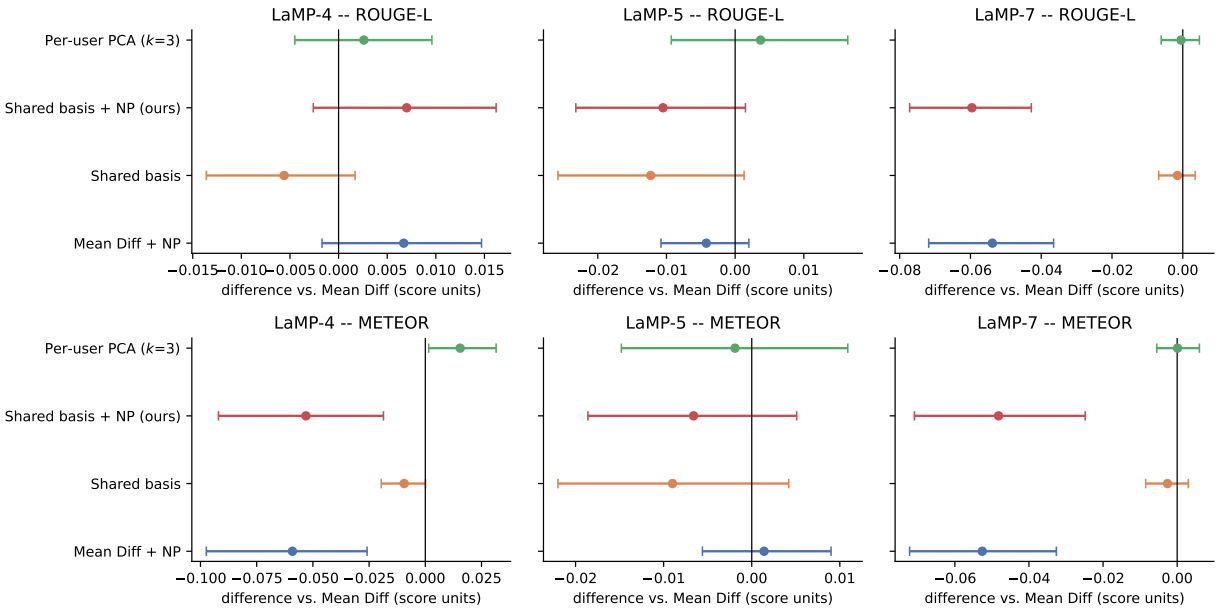


Figure 7: Run 5 paired differences against Mean Difference with 95% intervals. Points to the right of zero are better than the paper's method.

Question (Run 5, paired vs. Mean Diff)	Comparisons	Sig. better	Sig. worse	Inconclusive
Norm preservation on the single vector	6	0	3	3
Shared-basis multi-component vs. single vector	6	0	0	6
Full method (shared basis + NP) vs. paper method	6	0	3	3
Per-user PCA (k=3) vs. single vector	6	1	0	5

Table 20: Scorecard of the Run 5 paired comparisons (ROUGE-L and METEOR, three tasks).

8.5 Findings

- **The paper's method works in the refined pipeline.** Unlike Runs 2–4, Mean Difference gives gains of the reported size on LaMP-5 and on LaMP-7 ROUGE-L. Because the pipeline changes were introduced together, this run does not isolate which change (cleaning, history construction, matched prompts, all-position steering or per-condition tuning) accounts for the improvement.
- **The shared-basis vector matches Mean Difference.** It is inconclusive in all six comparisons, with differences of at most about 0.012 in absolute score. On LaMP-7 the difference is essentially zero with a tight interval (-0.0015, $[-0.0068, +0.0035]$). This is consistent with the Run 2 diagnostic that 95% of LaMP-7 user vectors are nearly identical to the global vector: projecting a vector that lies mostly along a shared direction onto a basis that contains that direction changes little. On LaMP-4 and LaMP-5 the point estimates are slightly lower (ROUGE-L -0.006 and -0.012) but not significant.
- **Norm-preserving variants score lower than Mean Difference.** Both NP conditions are significantly lower in three of six comparisons each (LaMP-7 in both metrics and LaMP-4 METEOR). The selected strengths (≤ 1.0) and the tuning-user scores show the same ordering, so larger strengths were not preferred. On LaMP-7 NP returns ROUGE-L to the unsteered level. A possible explanation, not tested here, is that the LaMP-7 effect relies on changing the activation magnitude, which NP removes by construction. On LaMP-4 the combination of slightly higher ROUGE-L and much lower METEOR is the pattern expected if NP outputs are shorter, since METEOR is recall-weighted. Output lengths would confirm this.
- **Per-user PCA matches Mean Difference.** With tuned strength in the refined pipeline, per-user PCA (k=3) equals Mean Difference in five of six comparisons and is better on LaMP-4 METEOR ($+0.0155$). It is not significantly better on ROUGE-L or on the other tasks.
- **Ablation outcome.** The full method (shared basis + NP) is not significantly better than the paper's method on any task, and is lower on LaMP-7 and on LaMP-4 METEOR. Mean Difference steering with tuned layer and strength is the strongest configuration tested.

9 Cross-run synthesis

9.1 The 2×2 ablation across runs

Tables 21 and 22 gather the proposal's ablation grid for every run. Run 3 contains only the multi-component columns, and for Run 5 "Multi" is the shared-basis vector. Absolute levels differ between runs because of the number and sample of users, cleaning, and prompt construction (compare the baseline columns), so the *relative* comparisons in the remainder of this section are the meaningful ones. Run 2's multi-component column uses k tuned per task ($3/5/3$); the k used in Run 3 is not recorded.

Task	Run	No steering	Single, add	Single + NP	Multi, add	Multi + NP
LaMP-4	1	0.0909	0.0907	0.0909	0.0886	0.0895
	2	0.1105	0.1156	0.1136	0.1136	0.1127
	3	0.1098	–	–	0.1132	0.1123
	4	0.0971	0.0963	0.0963	0.0961	0.0959
	5	0.0569	0.0609	0.0676	0.0553	0.0679
LaMP-5	1	0.2137	0.2203	0.2222	0.2261	0.2271
	2	0.3240	0.3229	0.3199	0.3218	0.3213
	3	0.3196	–	–	0.3205	0.3197
	4	0.2118	0.2106	0.2104	0.2089	0.2095
	5	0.2578	0.3141	0.3100	0.3019	0.3036
LaMP-7	1	0.3584	0.3577	0.3554	0.3579	0.3535
	2	0.2975	0.3016	0.2978	0.3042	0.3035
	3	0.2938	–	–	0.2975	0.2988
	4	0.2429	0.3133	0.3041	0.3132	0.3081
	5	0.3001	0.3527	0.2989	0.3512	0.2931

Table 21: ROUGE-L for the 2×2 grid, by run.

Task	Run	No steering	Single, add	Single + NP	Multi, add	Multi + NP
LaMP-4	1	0.1203	0.1155	0.1162	0.1143	0.1149
	2	0.0933	0.1006	0.1001	0.0981	0.0983
	3	0.0966	–	–	0.1027	0.1020
	4	0.1206	0.1207	0.1202	0.1200	0.1198
	5	0.0574	0.1436	0.0844	0.1341	0.0905
LaMP-5	1	0.3179	0.3131	0.3160	0.3284	0.3300
	2	0.3269	0.3246	0.3220	0.3252	0.3235
	3	0.3348	–	–	0.3346	0.3340
	4	0.3122	0.3116	0.3116	0.3095	0.3107
	5	0.3076	0.3411	0.3425	0.3321	0.3345
LaMP-7	1	0.3440	0.3317	0.3296	0.3273	0.3210
	2	0.2640	0.2746	0.2716	0.2754	0.2759
	3	0.2618	–	–	0.2668	0.2678
	4	0.2644	0.3100	0.2977	0.3086	0.3051
	5	0.4055	0.4147	0.3620	0.4121	0.3665

Table 22: METEOR for the 2×2 grid, by run.

9.2 Effect of the paper's method and of the proposed extensions

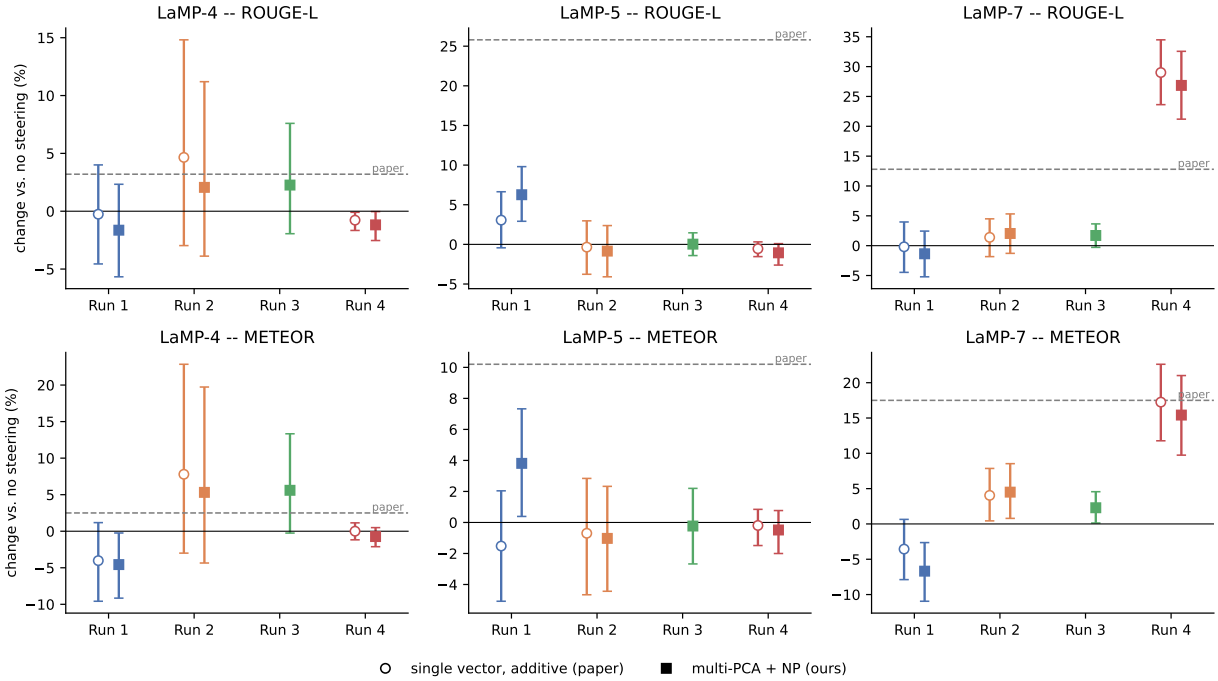


Figure 8: Change vs. no steering for the paper's method (open circles) and the full proposed method (filled squares), by run, with 95% intervals (Runs 1–4; Run 5 has no intervals against no steering, see Figure 7). Dashed grey lines: the paper's reported relative improvements (Table 2 of the paper). Run 4 is scored on raw outputs, so its LaMP-7 effect includes the formatting artefact discussed in Section 7.

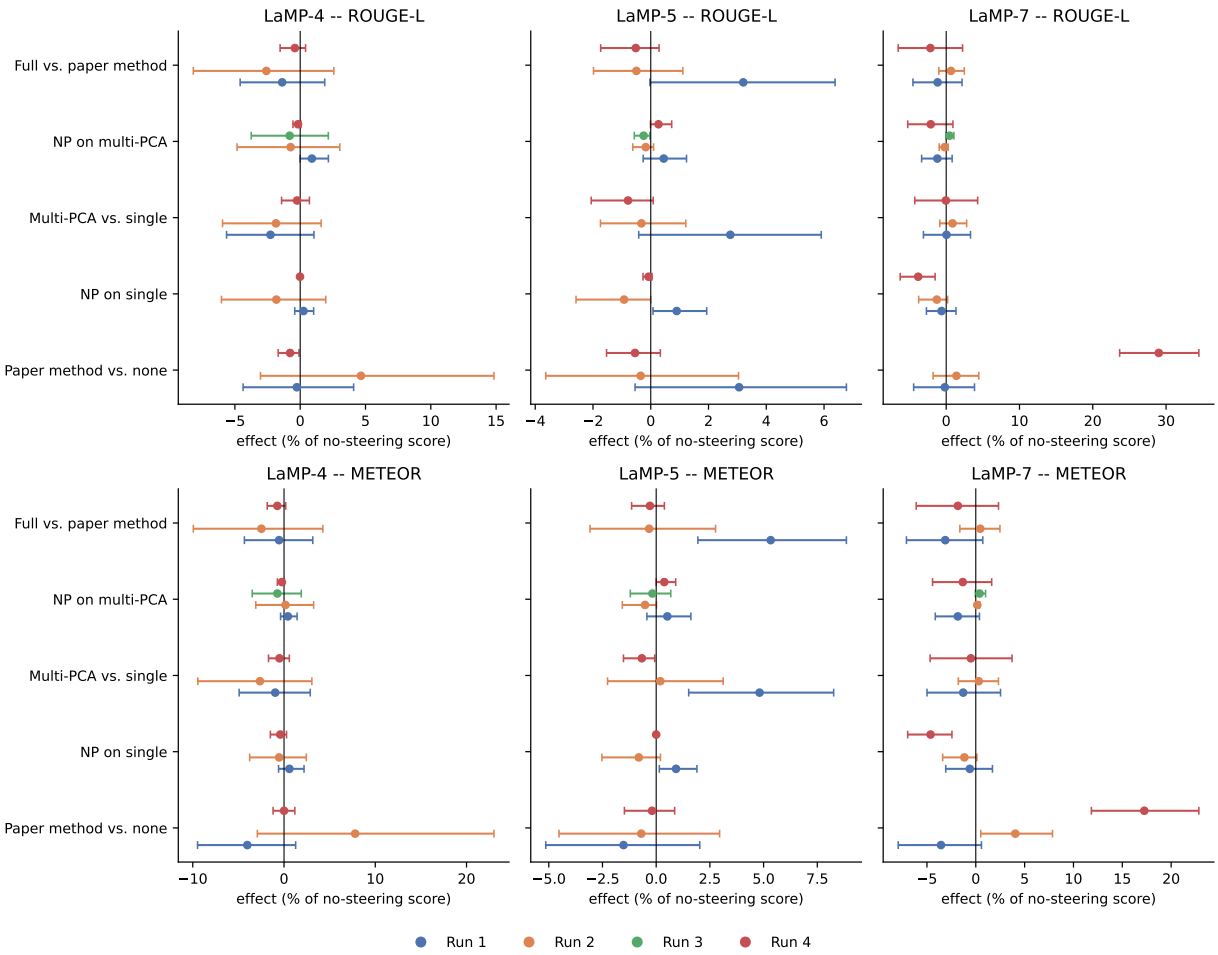


Figure 9: Paired effects of each mechanism (95% intervals, as % of the no-steering score) in Runs 1–4. Rows with no marker for a run were not evaluated in that run.

Question	Comparisons	Sig. better	Sig. worse	Inconclusive
Paper method (single, additive) vs. no steering	18	3	1	14
Norm preservation on the single vector	18	2	2	14
Multi-PCA vs. single vector (additive)	18	1	1	16
Norm preservation on multi-PCA	24	0	1	23
Full method (multi-PCA + NP) vs. paper method	18	1	0	17
Full method vs. no steering	24	6	3	15

Table 23: Scorecard over all paired comparisons in Runs 1–4 (ROUGE-L and METEOR, three tasks, all runs where the comparison exists). Run 5 is summarised in Table 20.

The scorecard (Table 23) summarises the evidence:

- **Does StyleVector help?** Three of 18 comparisons are significantly positive (Run 2 LaMP-7 METEOR, and Run 4 LaMP-7 ROUGE-L and METEOR), one is significantly negative (Run 4 LaMP-4 ROUGE-L) and 14 are inconclusive.
- **Does multi-component PCA help?** One of 18 comparisons is significantly positive (Run 1 LaMP-5 METEOR) and one significantly negative (Run 4 LaMP-5 METEOR). The positive result is not replicated by Runs 2–4 on the same task.
- **Does NP help?** Of 42 comparisons of NP against the corresponding additive variant, 2 are significantly better (both Run 1 LaMP-5, about +0.9%), 3 significantly worse (Run 3 LaMP-5, and Run 4 LaMP-7)

and 37 are inconclusive.

- **Does the full method beat the paper's method?** Not significantly in 17 of 18 comparisons. The exception is Run 1 LaMP-5 METEOR (+5.3%).

Run 5 adds 24 paired comparisons against Mean Difference (Table 20): the shared basis is never different, NP on the single vector is significantly lower in 3 of 6, the full method is significantly lower in 3 of 6, and per-user PCA is better in 1 of 6. In the proposal, success was defined as a measurable improvement of the full method over the reproduced single-vector baseline on at least two of three tasks. By this criterion the hypothesis is **not supported** by Runs 1–5.

9.3 Comparison with the paper

Task	Source	ROUGE-L Δ%	[95% CI]	METEOR Δ%	[95% CI]
LaMP-4	Paper (StyleVector, Table 2)	+3.2	–	+2.5	–
	Run 1 – single	-0.3	[-4.6,+4.0]	-4.0	[-9.6,+1.2]
	Run 2 – single	+4.7	[-3.0,+14.8]	+7.8	[-3.0,+22.8]
	Run 3 – PCA, additive	+3.1	[-2.0,+8.8]	+6.3	[-0.1,+14.2]
	Run 4 – single	-0.8	[-1.7,-0.1]	+0.0	[-1.2,+1.1]
	Run 5 – single	+7.0	n/a	+150.2	n/a
LaMP-5	Paper (StyleVector, Table 2)	+25.8	–	+10.2	–
	Run 1 – single	+3.1	[-0.4,+6.6]	-1.5	[-5.1,+2.0]
	Run 2 – single	-0.3	[-3.8,+3.0]	-0.7	[-4.7,+2.8]
	Run 3 – PCA, additive	+0.3	[-1.1,+1.7]	-0.1	[-2.3,+2.2]
	Run 4 – single	-0.5	[-1.5,+0.3]	-0.2	[-1.5,+0.8]
	Run 5 – single	+21.8	n/a	+10.9	n/a
LaMP-7	Paper (StyleVector, Table 2)	+12.8	–	+17.5	–
	Run 1 – single	-0.2	[-4.5,+4.0]	-3.6	[-7.9,+0.6]
	Run 2 – single	+1.4	[-1.8,+4.5]	+4.0	[+0.4,+7.9]
	Run 3 – PCA, additive	+1.3	[-0.7,+3.2]	+1.9	[-0.2,+4.1]
	Run 4 – single	+29.0	[+23.6,+34.5]	+17.2	[+11.8,+22.6]
	Run 5 – single	+17.5	n/a	+2.3	n/a

Table 24: Relative improvement of the paper's method over its own non-personalized baseline vs. the reproductions. Run 4 is scored on raw outputs. Run 5 values are computed from its absolute means; no intervals against no steering are available. After cleaning, the LaMP-7 ROUGE-L gain is +12.5% (Table 13). The paper's numbers are on its own split and protocol.

In Runs 1–4 only LaMP-7 approaches the paper's magnitude (Run 4: +12.5% ROUGE-L after cleaning vs. the paper's +12.8%), and that effect is likely not user-specific. The paper's large LaMP-5 gain (+25.8% ROUGE-L) is not reproduced in Runs 1–4: those runs range from −0.5% to +3.1% with intervals including zero. Run 5 reproduces it (+21.8% ROUGE-L, and +10.9% METEOR against +10.2%), which coincides with the pipeline refinements. Run 5 also improves ROUGE-L on LaMP-4 (+7.0% against +3.2%) and LaMP-7 (+17.5% against +12.8%). METEOR is higher than the paper's on LaMP-4 (from a low unsteered baseline) and lower on LaMP-7 (+2.3% against +17.5%). Absolute levels differ between runs because of different users, prompts and cleaning, so each run's gain is measured against its own baseline.

9.4 The output-format confound

Task	Run	Raw outputs	No steering	Single, add	Multi + NP
LaMP-4	1	as generated	93.0	94.7	94.0
	2	before cleaning	97.3	97.3	97.3
	3	before cleaning	96.3	–	96.3
	4	as generated	97.7	97.7	97.7
LaMP-5	1	as generated	99.7	99.0	99.3
	2	before cleaning	100.0	100.0	100.0
	3	before cleaning	100.0	–	100.0
	4	as generated	100.0	100.0	100.0
LaMP-7	1	as generated	17.0	3.3	3.0
	2	before cleaning	86.0	85.3	85.3
	3	before cleaning	83.0	–	82.7
	4	as generated	84.9	21.5	21.5

Table 25: Percentage of raw outputs beginning with “Sure”, “Here”, “Certainly”, etc. For Runs 2 and 3 the raw field is used (before cleaning); Run 3 has no single-vector condition, so “Single, add” is blank.

Task	Run	ROUGE-L, raw output	ROUGE-L, cleaned	Relative gain from cleaning (%)
LaMP-4	2	0.0941	0.1105	+17.5
	3	0.0960	0.1098	+14.5
LaMP-5	2	0.2152	0.3240	+50.6
	3	0.2113	0.3196	+51.3
LaMP-7	2	0.2467	0.2975	+20.6
	3	0.2470	0.2938	+18.9

Table 26: Effect of output cleaning on the *unsteered* baseline ROUGE-L (Runs 2 and 3, per-user scores before and after cleaning).

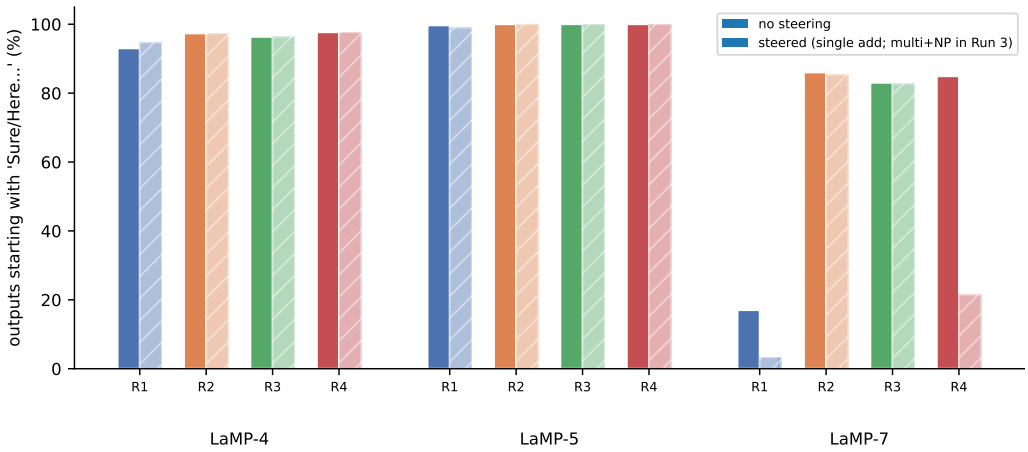


Figure 10: Share of chatty-prefix outputs without steering (solid) and with steering (hatched), by run.

Three observations follow. (i) For LaMP-4 and LaMP-5, 93–100% of raw generations start with a preamble in every run, and steering at the selected strengths does not change this. Cleaning alone raises baseline ROUGE-L by 14–18% (LaMP-4), 51% (LaMP-5) and 19–21% (LaMP-7), which is far larger than any steering effect observed in Runs 1–4. Run 1 and Run 4 scores therefore mix steering effects with formatting noise. (ii) On LaMP-7, strong steering (Run 4, $\alpha = 0.5$) reduces chatty outputs from 85% to 21%, while weak steering (Run 2, $\rho = 0.05$) leaves the rate unchanged (86% $\rightarrow$ 85%). (iii) Consequently, evaluating

on raw output can create or hide effects, and any final comparison must apply identical cleaning to all conditions.

9.5 Is the gain personal?

Three pieces of evidence indicate that the gains observed so far are largely not user-specific. First, the global vector is statistically indistinguishable from the user vector in Run 2 on all tasks (numerically higher on LaMP-7). Second, a random direction with the same strength reaches similar scores to PCA + NPin Run 3. Third, on LaMP-7 and for more than half of the LaMP-4 users, user vectors are nearly identical to the global vector (Table 7). On LaMP-5, where user vectors differ most from the global vector, steering showed no gain in Runs 2–4. In Run 5 there is no global-vector control, but the shared-basis projection (which discards user-specific variation outside the shared subspace) is no worse than Mean Difference, which again indicates that what steering contributes is largely shared across users.

9.6 Strength selection and the role of NP

Selection on validation users chose the smallest strength in the grid in all 24 selections (Runs 2 and 3), i.e. validation favoured small perturbations. It also means NP was evaluated in the regime where it has little room to act: to first order the rescaling in Eq. (2) removes only the component of p parallel to h, so at $\rho = 0.05$ the two variants differ by at most $0.05\|h\|$, and empirically they are nearly identical there (Figure 4). Where strength is large (Run 3 $\rho \geq 0.8$; Run 4 sweep) quality has already collapsed, and NP rescues it only partially. Run 5 tests this premise most directly, because NP conditions were tuned over larger strengths (α up to 4) on separate tuning users. The selected strengths were at most 1.0, and NP scored lower than the additive vector in 3 of 6 comparisons. In these experiments, then, NP did not permit larger, more effective steering strengths.

10 Discussion

Answers to the project's questions.

1. *Reproduce the paper's gains?* Yes in Run 5: LaMP-5 ROUGE-L and METEOR, and ROUGE-L on LaMP-4 and LaMP-7. Runs 1–4 reproduced them only partially (LaMP-7).
2. *Improve with multi-component vectors?* Per-user PCA (Runs 1–4) gave no consistent improvement. The shared-basis vector (Run 5) is indistinguishable from Mean Difference in 6 of 6 comparisons.
3. *Improve with norm preservation?* Runs 1–4: 2 of 42 comparisons significantly better. Run 5: lower than Mean Difference in 3 of 6 comparisons for each NP condition, even with larger tuned strengths.
4. *Ablation of the two components.* Complete for 2×2 (Tables 21–22), plus $k \in \{1,3,5\}$ (Run 4), layer and strength sweeps (Runs 2–4), and the shared-basis variant (Run 5).
5. *Qualitative analysis.* Table 14 shows LaMP-7 examples.

Limitations.

- **Evaluation noise.** Short targets (headlines, titles, tweets) and 150–300 users give wide intervals. Many comparisons are underpowered to detect effects of 1–3%.
- **Cleaning is a heuristic.** It was applied identically across conditions within a run, but it differs between runs, and runs are not directly comparable in absolute terms.
- **Duplicate profiles.** The Run 2 diagnostic indicates repeated users on LaMP-4, which reduces the effective sample size there.
- **Dev split, not the paper's test split.** The paper's absolute numbers are not comparable.

- **Recorded settings.** The model identifier, and for some runs the history cap and PCA k, are not stored in the checkpoint archives and are marked "not recorded".
- **Run 5 summary data.** Run 5 is reported from the notebook's summary tables, so intervals are available for the comparisons against Mean Difference but not against no steering.
- **Combined pipeline changes.** Run 5 changed several things at once, so the improvement from Runs 2–4 to Run 5 cannot be attributed to a single change.
- **No multiple-comparison correction** (Section 1).

What the early runs taught. Run 1 showed that fixed, reused hyper-parameters and raw outputs cannot support conclusions. Run 2 introduced the controls that revealed the weak personal signal. Run 3 confirmed that selected strengths sit at the grid edge and that a random direction is a competitive control. Run 4 showed that the only large effect is on LaMP-7 and is partly a format artefact. Run 5 showed that, with the refined pipeline, the paper's own method is the strongest configuration, and that neither extension improves on it.

11 Conclusion

Across five runs, the paper's mean-difference steering is the strongest configuration tested. In the refined Run 5 pipeline it reproduces the paper's gains on LaMP-5 and improves ROUGE-L on LaMP-4 and LaMP-7. Neither proposed extension improved on it: the shared-basis multi-component vector is statistically indistinguishable from Mean Difference, and norm-preserving variants score lower in half of the comparisons, with selected strengths staying at or below 1.0 even when larger values were available. The earlier runs contribute the supporting analysis: output cleaning is a first-order effect, user vectors on two of three tasks are close to the global vector, and a random direction is a competitive control. Together, the 2×2 ablation, the sweeps over k, layer and strength, and the control experiments give a complete account of which components of the proposal contribute to personalised generation.

A Paired comparisons for every run

Each table lists the paired effects (as % of the no-steering score) with 95% intervals. Comparisons that do not exist in a run are omitted.

Task	Comparison	ROUGE-L: Δ% of base [95% CI]	METEOR: Δ% of base [95% CI]
LaMP-4	Paper method vs. no steering	-0.3 [-4.4,+4.1]	-4.0 [-9.5,+1.3]
	NP alone (single vector)	+0.3 [-0.4,+1.0]	+0.6 [-0.6,+2.2]
	Multi-PCA alone vs. single	-2.3 [-5.6,+1.1]	-1.0 [-4.9,+2.9]
	NP on top of multi-PCA	+0.9 [-0.0,+2.2]	+0.4 [-0.4,+1.4]
	Full method vs. paper method	-1.4 [-4.6,+1.9]	-0.5 [-4.3,+3.2]
	Full method vs. no steering	-1.6 [-5.5,+2.4]	**-4.6 [-9.1,-0.1]**
LaMP-5	Paper method vs. no steering	+3.1 [-0.5,+6.8]	-1.5 [-5.1,+2.0]
	NP alone (single vector)	**+0.9 [+0.1,+1.9]**	**+0.9 [+0.1,+1.9]**
	Multi-PCA alone vs. single	+2.8 [-0.4,+5.9]	**+4.8 [+1.5,+8.3]**
	NP on top of multi-PCA	+0.4 [-0.3,+1.2]	+0.5 [-0.4,+1.6]
	Full method vs. paper method	+3.2 [-0.0,+6.4]	**+5.3 [+1.9,+8.9]**
	Full method vs. no steering	**+6.3 [+2.9,+9.8]**	**+3.8 [+0.4,+7.3]**
LaMP-7	Paper method vs. no steering	-0.2 [-4.4,+3.9]	-3.6 [-7.9,+0.6]
	NP alone (single vector)	-0.6 [-2.7,+1.3]	-0.6 [-3.1,+1.7]
	Multi-PCA alone vs. single	+0.0 [-3.1,+3.3]	-1.3 [-5.0,+2.5]
	NP on top of multi-PCA	-1.2 [-3.4,+0.8]	-1.8 [-4.1,+0.4]
	Full method vs. paper method	-1.2 [-4.5,+2.2]	-3.1 [-7.1,+0.7]
	Full method vs. no steering	-1.4 [-5.2,+2.5]	**-6.7 [-10.9,-2.6]**

Table 27: Run 1 paired comparisons.

Task	Comparison	ROUGE-L: Δ% of base [95% CI]	METEOR: Δ% of base [95% CI]
LaMP-4	Paper method vs. no steering	+4.7 [-3.0,+14.8]	+7.8 [-2.9,+23.0]
	NP alone (single vector)	-1.8 [-6.0,+2.0]	-0.5 [-3.8,+2.4]
	Multi-PCA alone vs. single	-1.9 [-5.9,+1.6]	-2.6 [-9.4,+3.1]
	NP on top of multi-PCA	-0.7 [-4.8,+3.0]	+0.2 [-3.1,+3.2]
	Full method vs. paper method	-2.6 [-8.2,+2.6]	-2.5 [-9.9,+4.3]
	Full method vs. no steering	+2.1 [-3.9,+11.2]	+5.3 [-4.4,+20.2]
LaMP-5	Paper method vs. no steering	-0.3 [-3.6,+3.0]	-0.7 [-4.5,+3.0]
	NP alone (single vector)	-0.9 [-2.6,+0.0]	-0.8 [-2.5,+0.2]
	Multi-PCA alone vs. single	-0.3 [-1.7,+1.2]	+0.2 [-2.3,+3.1]
	NP on top of multi-PCA	-0.2 [-0.6,+0.1]	-0.5 [-1.6,+0.0]
	Full method vs. paper method	-0.5 [-2.0,+1.1]	-0.3 [-3.1,+2.8]
	Full method vs. no steering	-0.8 [-4.1,+2.4]	-1.0 [-4.3,+2.4]
LaMP-7	Paper method vs. no steering	+1.4 [-1.8,+4.4]	**+4.0 [+0.5,+7.9]**
	NP alone (single vector)	-1.3 [-3.8,+0.2]	-1.2 [-3.4,+0.1]
	Multi-PCA alone vs. single	+0.9 [-0.9,+2.8]	+0.3 [-1.8,+2.3]
	NP on top of multi-PCA	-0.2 [-1.0,+0.3]	+0.2 [-0.0,+0.4]
	Full method vs. paper method	+0.6 [-1.0,+2.5]	+0.5 [-1.6,+2.5]
	Full method vs. no steering	+2.0 [-1.3,+5.3]	**+4.5 [+0.8,+8.4]**

Table 28: Run 2 paired comparisons.

Task	Comparison	ROUGE-L: Δ% of base [95% CI]	METEOR: Δ% of base [95% CI]
LaMP-4	NP on top of multi-PCA	-0.8 [-3.7,+2.2]	-0.7 [-3.5,+1.9]
	Full method vs. no steering	+2.3 [-1.9,+7.6]	+5.6 [-0.2,+13.2]
LaMP-5	NP on top of multi-PCA	**-0.2 [-0.6,-0.0]**	-0.2 [-1.2,+0.7]
	Full method vs. no steering	+0.0 [-1.5,+1.5]	-0.2 [-2.8,+2.2]
LaMP-7	NP on top of multi-PCA	+0.5 [+0.0,+1.1]	+0.4 [-0.0,+1.0]
	Full method vs. no steering	+1.7 [-0.2,+3.6]	**+2.3 [+0.1,+4.5]**

Table 29: Run 3 paired comparisons.

Task	Comparison	ROUGE-L: Δ% of base [95% CI]	METEOR: Δ% of base [95% CI]
LaMP-4	Paper method vs. no steering	**-0.8 [-1.7,-0.1]**	+0.0 [-1.2,+1.2]
	NP alone (single vector)	-0.0 [-0.0,+0.0]	-0.4 [-1.5,+0.3]
	Multi-PCA alone vs. single	-0.2 [-1.4,+0.7]	-0.5 [-1.7,+0.6]
	NP on top of multi-PCA	-0.2 [-0.6,+0.1]	-0.2 [-0.7,+0.0]
	Full method vs. paper method	-0.4 [-1.5,+0.4]	-0.7 [-1.8,+0.2]
	Full method vs. no steering	**-1.2 [-2.6,-0.0]**	-0.7 [-2.1,+0.5]
LaMP-5	Paper method vs. no steering	-0.5 [-1.5,+0.3]	-0.2 [-1.5,+0.9]
	NP alone (single vector)	-0.1 [-0.3,+0.0]	-0.0 [-0.0,+0.0]
	Multi-PCA alone vs. single	-0.8 [-2.1,+0.1]	**-0.7 [-1.5,-0.1]**
	NP on top of multi-PCA	+0.3 [-0.0,+0.7]	**+0.4 [+0.0,+0.9]**
	Full method vs. paper method	-0.5 [-1.7,+0.3]	-0.3 [-1.1,+0.4]
	Full method vs. no steering	-1.1 [-2.6,+0.1]	-0.5 [-2.0,+0.8]
LaMP-7	Paper method vs. no steering	**+29.0 [+23.7,+34.5]**	**+17.2 [+11.8,+22.8]**
	NP alone (single vector)	**-3.8 [-6.3,-1.5]**	**-4.6 [-7.0,-2.4]**
	Multi-PCA alone vs. single	-0.0 [-4.3,+4.3]	-0.5 [-4.7,+3.7]
	NP on top of multi-PCA	-2.1 [-5.2,+0.9]	-1.3 [-4.4,+1.6]
	Full method vs. paper method	-2.2 [-6.5,+2.2]	-1.8 [-6.1,+2.3]
	Full method vs. no steering	**+26.9 [+21.3,+32.5]**	**+15.4 [+9.8,+21.0]**

Table 30: Run 4 paired comparisons (raw outputs).

B Additional original figures

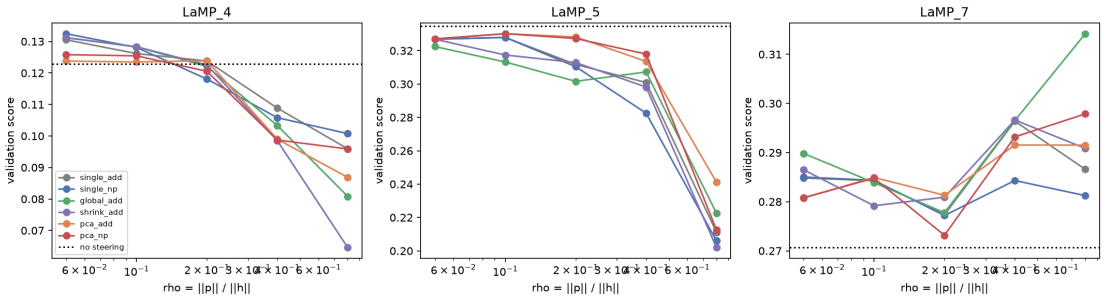


Figure 11: Run 2 (original): validation score vs. ρ.

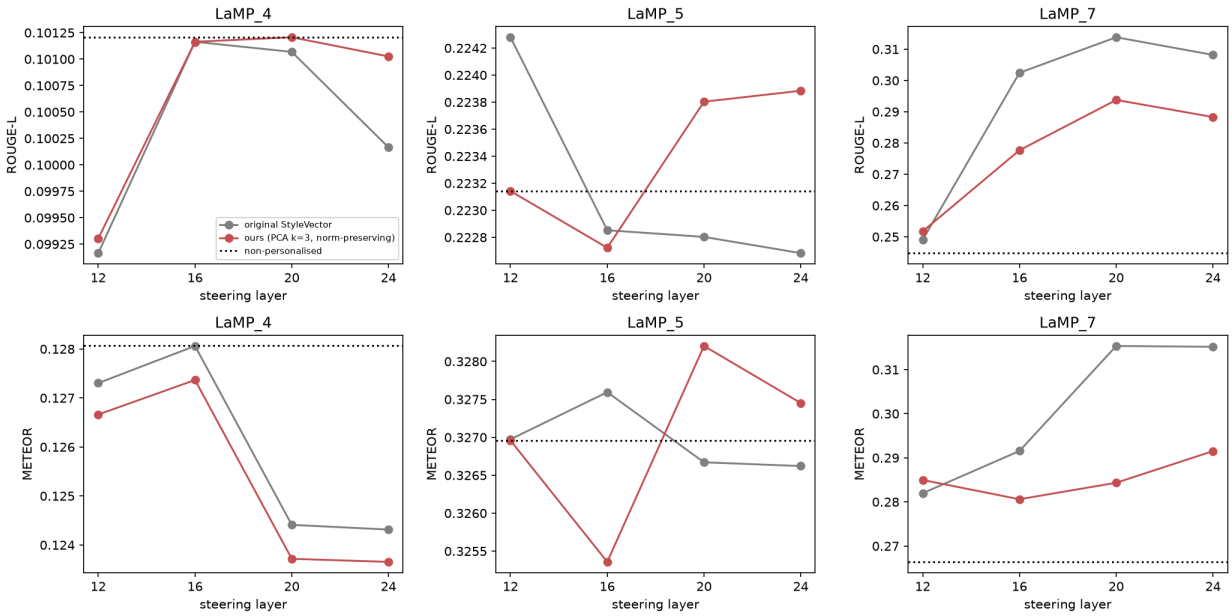


Figure 12: Run 4 (original): layer sweep.

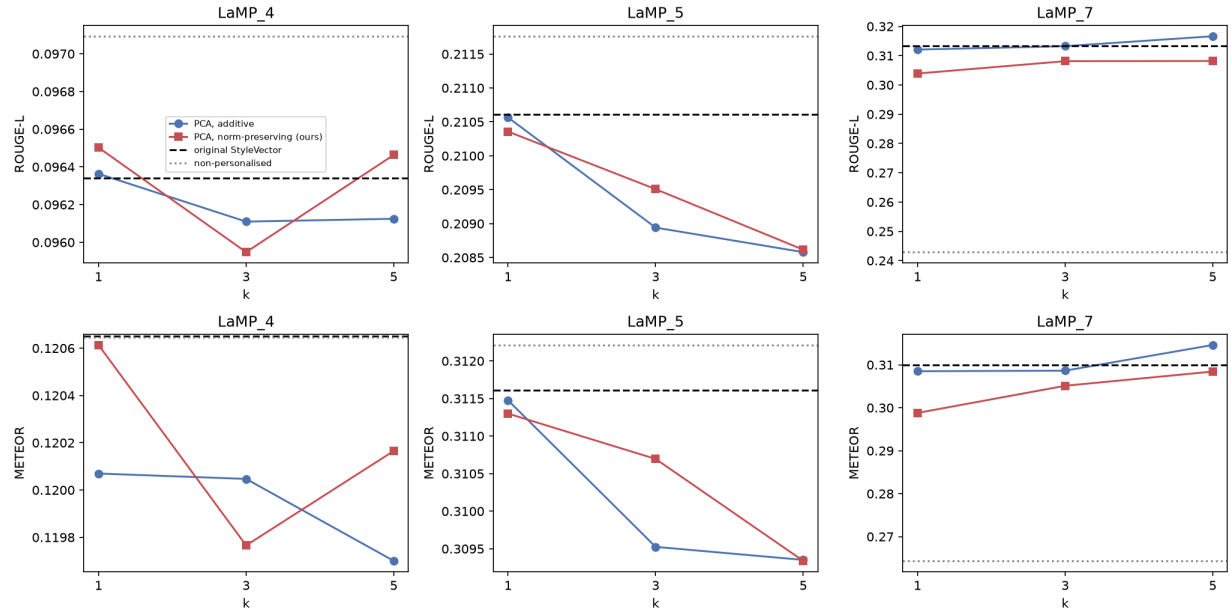


Figure 13: Run 4 (original): sweep over the number of PCA components k.

C Use of AI assistance

An AI assistant (Anthropic's Claude) was used to review the earlier notebook, to propose the corrected pipeline used in Run 5 (Section 8), and to draft the analysis and LaTeX for this report from the saved checkpoint files and the Run 5 summary output. All experiments in Runs 1–5 were run by the author. The statistical recomputation is reproducible from the archives with the scripts supplied alongside this document.